

Multispecies Boltzmann Equation

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Abstract

These notes on the multi-species Boltzmann equation addresses gaps in the literature and offers a modern, pedagogical introduction to this fundamental kinetic framework.

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1 Introduction

Our primary objective in this document is to model the interaction of dilute mixtures of particles (gas mixtures). The most accurate kinetic model for such mixtures is a multi-species Boltzmann system, which we will now try to understand. Typically, the direct approximation of multi-species Boltzmann equations is intractable. We will ultimately turn to BGK-type approximations of the Boltzmann models, but we must understand some aspects of the Boltzmann model in order to effectively design and use the approximations.

In compiling the material for this chapter, we have found the thesis by Marlies Pirner [7] especially useful. The books [3, 4] also have short, readable descriptions of the multi-species Boltzmann system. The book by Kremer [5] has a fairly in-depth and modern account of the multi-species Boltzmann equation, and the notes by Alberto Bressan [2] were helpful in understanding the basics of the unary Boltzmann model as well as its mathematical analysis.

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2 Quick Introduction to the Unary Boltzmann Equation

The *Vlasov–Boltzmann equation* for a single-species dilute gas is given as follows:

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f + \mathbf{a} \cdot \nabla_{\mathbf{v}} f = Q[f, f](\mathbf{x}, \mathbf{v}, t), \quad (1)$$

where f is the density of particles at position $\mathbf{x}$ with velocity $\mathbf{v}$ at time t ; and $\mathbf{a}$ is a particle acceleration determined by an external field, for example, an electric or magnetic field. The *Boltzmann transport equation* results from setting $\mathbf{a} \equiv \mathbf{0}$. The derivation of the equation is a straightforward exercise using the chain rule. In particular, the total time derivative of the distribution f can be realized as

$$\begin{aligned} \frac{d}{dt} f(\mathbf{x}, \mathbf{v}, t) &= \sum_{i=1}^d \frac{\partial f}{\partial x_i} \frac{dx_i}{dt} + \sum_{i=1}^d \frac{\partial f}{\partial v_i} \frac{dv_i}{dt} + \frac{\partial f}{\partial t} \\ &= \mathbf{v} \cdot \nabla_{\mathbf{x}} f + \mathbf{a} \cdot \nabla_{\mathbf{v}} f + \frac{\partial f}{\partial t}, \end{aligned}$$

where $d = 3$ for the standard model. Thus, the Vlasov–Boltzmann equation may be constructed via the following law: the total time rate of change of the distribution is equal to the collision (frequency) operator. In other words,

$$\frac{d}{dt} f(\mathbf{x}, \mathbf{v}, t) = Q[f, f](\mathbf{x}, \mathbf{v}, t).$$

The *Boltzmann collision operator*, $Q[f, f]$, requires much more physical insight for a clear derivation [8], involving certain conservation principles, in particular. Therefore, we will content ourselves by only stating its generic form in three dimensions for the moment:

$$Q[f, f](\mathbf{x}, \mathbf{v}, t) = \frac{1}{m} \int_{\mathbb{R}^d} \int_{\mathbb{S}^{d-1}} [f' f'_* - f f_*] B(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) d\boldsymbol{\sigma} d\mathbf{v}_*,$$

where

$$\begin{aligned} f' &:= f(\mathbf{x}, \mathbf{v}', t), & f'_* &:= f(\mathbf{x}, \mathbf{v}'_*, t), \\ f &:= f(\mathbf{x}, \mathbf{v}, t), & f_* &:= f(\mathbf{x}, \mathbf{v}_*, t); \end{aligned}$$

m is the mass of an individual particle; B is the collision kernel describing binary collisions between particles; $\boldsymbol{\sigma}$ is a unit vector called the apse direction; and the primed coordinates are

computed via the transformation

$$\mathbf{v}' = \mathbf{v} - \boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*), \quad (2)$$

$$\mathbf{v}'_* = \mathbf{v}_* + \boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*). \quad (3)$$

The computation of the Boltzmann collision operator is very expensive. For the 3D case, the (5 dimensional) integral must be computed at every value of $(\mathbf{x}, \mathbf{v})$ in phase space. Additionally, the integral cannot be evaluated analytically, except for the simplest of cases (e.g., Maxwell molecules, with carefully prepared initial conditions). Thus, computation of the collision operator is usually the most expensive part of computing numerical solutions of the Boltzmann equation. For this reason, Monte Carlo (MC) methods are generally preferred over direct numerical simulation [6]. Unfortunately, MC methods are slow to converge, as is well known. What is worse, for the N -species case, there are N^2 collision operators that must be considered. Thus, a simpler, less costly BGK-type approximation is often implemented. BGK stands for the names Bhatnagar, Gross, and Krook, who introduced the approximation in 1954 [1], and we now explain this methodology.

3 The BGK Approximation Model

The *Vlasov–BGK equation* is a model derived by approximating the Boltzmann collision operator with a simpler nonlinear, nonlocal relaxation operator of the form

$$Q^{BGK}[f](\mathbf{x}, \mathbf{v}, t) = \lambda(M_f(\mathbf{x}, \mathbf{v}, t) - f(\mathbf{x}, \mathbf{v}, t)), \quad (4)$$

where $\lambda = \frac{1}{\tau}$ is the collision frequency between particles, $\tau > 0$ is a characteristic time, and M_f is the *Maxwellian*, which is constructed using the first three integral moments of f ,

$$n(\mathbf{x}, t) := \int_{\mathbb{R}^d} f \, d\mathbf{v}, \quad \mathbf{u}(\mathbf{x}, t) := \frac{1}{n} \int_{\mathbb{R}^d} f \mathbf{v} \, d\mathbf{v}, \quad \theta(\mathbf{x}, t) := \frac{m}{nd} \int_{\mathbb{R}^d} f |\mathbf{v} - \mathbf{u}|^2 \, d\mathbf{v}, \quad (5)$$

and is defined as

$$M_f(\mathbf{x}, \mathbf{v}, t) := M[f](\mathbf{x}, \mathbf{v}, t) := n \left(\frac{m}{2\pi\theta} \right)^{d/2} \exp \left(-\frac{m|\mathbf{v} - \mathbf{u}|^2}{2\theta} \right). \quad (6)$$

The space-time fields n , $\mathbf{u}$, and θ , are the macroscopic (coarse grained) number density, bulk velocity, and temperature, respectively. The following macroscopic variables are also of

interest: the (kinetic) energy density is given by

$$\omega(\mathbf{x}, t) := \frac{m}{2} \int_{\mathbb{R}^d} f |\mathbf{v}|^2 d\mathbf{v}, \quad (7)$$

and the internal energy density is defined as

$$e(\mathbf{x}, t) := \frac{m}{2} \int_{\mathbb{R}^d} f |\mathbf{v} - \mathbf{u}|^2 d\mathbf{v}. \quad (8)$$

We observe that the temperature and the internal energy are related via

$$\theta(\mathbf{x}, t) = \frac{2}{d} \frac{e(\mathbf{x}, t)}{n(\mathbf{x}, t)}.$$

4 The Binary Collision Transformation

To derive the multispecies Boltzmann equation, we need some machinery.

Definition 4.1. Let $m_i, m_j > 0$ be the masses of particles of species i and j , respectively, $i, j \in \{1, 2, \dots, N\}$. Fix a unit vector $\boldsymbol{\sigma} \in \mathbb{S}^2$. The transformation $T_{\boldsymbol{\sigma}}^{ij}(\mathbf{v}, \mathbf{v}_*) = (\mathbf{v}', \mathbf{v}'_*)$, which maps $\mathbb{R}^3 \times \mathbb{R}^3$ into $\mathbb{R}^3 \times \mathbb{R}^3$, is defined via

$$\mathbf{v}' = \mathbf{v} - [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_j}{m_i + m_j} \boldsymbol{\sigma}, \quad (9)$$

$$\mathbf{v}'_* = \mathbf{v}_* + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_i}{m_i + m_j} \boldsymbol{\sigma}. \quad (10)$$

We will write

$$\mathbf{v}' = T_{\boldsymbol{\sigma},1}^{ij}(\mathbf{v}, \mathbf{v}_*) \quad \text{and} \quad \mathbf{v}'_* = T_{\boldsymbol{\sigma},2}^{ij}(\mathbf{v}, \mathbf{v}_*)$$

for the first and second components of the transformation, respectively. The mapping $T_{\boldsymbol{\sigma}}^{ij}$ is called a **binary collision transformation** and $\boldsymbol{\sigma}$ is called the **apse direction**. The un-primed coordinates $\mathbf{v}$ and $\mathbf{v}_*$ are called the **pre-collision velocities** and the primed coordinates $\mathbf{v}'$ and $\mathbf{v}'_*$ are called the **post-collision velocities**.

Proposition 4.2. Let $m_i, m_j > 0$ be the masses of particles of species i and j , respectively, $i, j \in \{1, 2, \dots, N\}$, and fix an apse direction $\boldsymbol{\sigma} \in \mathbb{S}^2$. Suppose that $(\mathbf{v}', \mathbf{v}'_*) = T_{\boldsymbol{\sigma}}^{ij}(\mathbf{v}, \mathbf{v}_*)$. Then,

$$m_i \mathbf{v} + m_j \mathbf{v}_* = m_i \mathbf{v}' + m_j \mathbf{v}'_*, \quad (11)$$

$$\frac{m_i}{2} |\mathbf{v}|^2 + \frac{m_j}{2} |\mathbf{v}_*|^2 = \frac{m_i}{2} |\mathbf{v}'|^2 + \frac{m_j}{2} |\mathbf{v}'_*|^2. \quad (12)$$

Furthermore,

$$\boldsymbol{\sigma} \cdot (\mathbf{v}' - \mathbf{v}'_*) = -\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*) \quad (13)$$

and

$$|\mathbf{v}' - \mathbf{v}'_*| = |\mathbf{v} - \mathbf{v}_*|. \quad (14)$$

Proof. To get (11) add m_i times (9) to m_j times (10), and the result immediately follows. To get (12), square both (9) and (10) to obtain

$$\begin{aligned} |\mathbf{v}'|^2 &= |\mathbf{v}|^2 - [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{4m_j}{m_i + m_j} \boldsymbol{\sigma} \cdot \mathbf{v} + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)]^2 \frac{4m_j^2}{(m_i + m_j)^2}, \\ |\mathbf{v}'_*|^2 &= |\mathbf{v}_*|^2 + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{4m_i}{m_i + m_j} \boldsymbol{\sigma} \cdot \mathbf{v}_* + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)]^2 \frac{4m_i^2}{(m_i + m_j)^2}. \end{aligned}$$

Then,

$$\begin{aligned} m_i |\mathbf{v}'|^2 + m_j |\mathbf{v}'_*|^2 &= m_i |\mathbf{v}|^2 + m_j |\mathbf{v}_*|^2 \\ &\quad - [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{4m_i m_j}{m_i + m_j} \boldsymbol{\sigma} \cdot \mathbf{v} + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{4m_i m_j}{m_i + m_j} \boldsymbol{\sigma} \cdot \mathbf{v}_* \\ &\quad + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)]^2 \frac{4m_i m_j^2}{(m_i + m_j)^2} + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)]^2 \frac{4m_i^2 m_j}{(m_i + m_j)^2} \\ &= m_i |\mathbf{v}|^2 + m_j |\mathbf{v}_*|^2 \\ &\quad - [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)]^2 \frac{4m_i m_j}{m_i + m_j} \\ &\quad + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)]^2 \frac{4m_i m_j}{(m_i + m_j)^2} (m_i + m_j) \\ &= m_i |\mathbf{v}|^2 + m_j |\mathbf{v}_*|^2. \end{aligned}$$

The proofs of identities (13) and (14) are left as simple exercises. ■

Proposition 4.3. *Let $m_i, m_j > 0$ be the masses of particles of species i and j , respectively, $i, j \in \{1, 2, \dots, N\}$, and fix an apse direction $\boldsymbol{\sigma} \in \mathbb{S}^2$. Then, $T_{\boldsymbol{\sigma}}^{ij}$ is a linear transformation, that is,*

$$T_{\boldsymbol{\sigma}}^{ij}(\alpha(\mathbf{v}, \mathbf{v}_*) + \beta(\mathbf{w}, \mathbf{w}_*)) = \alpha T_{\boldsymbol{\sigma}}^{ij}(\mathbf{v}, \mathbf{v}_*) + \beta T_{\boldsymbol{\sigma}}^{ij}(\mathbf{w}, \mathbf{w}_*), \quad (15)$$

for all $\alpha, \beta \in \mathbb{R}$ and all $\mathbf{v}, \mathbf{w}, \mathbf{v}_*, \mathbf{w}_* \in \mathbb{R}^3$. Furthermore,

$$(\mathbf{v}, \mathbf{v}_*) = T_{\boldsymbol{\sigma}}^{ij}(T_{\boldsymbol{\sigma}}^{ij}(\mathbf{v}, \mathbf{v}_*)), \quad (16)$$

for all $\mathbf{v}, \mathbf{v}_* \in \mathbb{R}^3$. In other words,

$$I = T_{\sigma}^{ij} \circ T_{\sigma}^{ij}, \quad (17)$$

or, equivalently,

$$(T_{\sigma}^{ij})^{-1} = T_{\sigma}^{ij}. \quad (18)$$

Proof. The linearity of T_{σ}^{ij} is a simple demonstration and is left to the reader. To prove the identity (16), observe that

$$\begin{aligned} & T_{\sigma,1}^{ij}(T_{\sigma}^{ij}(\mathbf{v}, \mathbf{v}_*)) \\ &= \mathbf{v} - [\sigma \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_j}{m_i + m_j} \sigma \\ &\quad - \left[\sigma \cdot \left(\mathbf{v} - [\sigma \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_j}{m_i + m_j} \sigma - \mathbf{v}_* - [\sigma \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_i}{m_i + m_j} \sigma \right) \right] \frac{2m_j}{m_i + m_j} \sigma \\ &= \mathbf{v} - [\sigma \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_j}{m_i + m_j} \sigma \\ &\quad - \left[\sigma \cdot \left((\mathbf{v} - \mathbf{v}_*) - [\sigma \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2(m_i + m_j)}{m_i + m_j} \sigma \right) \right] \frac{2m_j}{m_i + m_j} \sigma \\ &= \mathbf{v} - [\sigma \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_j}{m_i + m_j} \sigma \\ &\quad - [\sigma \cdot ((\mathbf{v} - \mathbf{v}_*) - 2[\sigma \cdot (\mathbf{v} - \mathbf{v}_*)] \sigma)] \frac{2m_j}{m_i + m_j} \sigma \\ &= \mathbf{v} - [\sigma \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_j}{m_i + m_j} \sigma + [\sigma \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_j}{m_i + m_j} \sigma \\ &= \mathbf{v}. \end{aligned}$$

A similar calculation shows that

$$T_{\sigma,2}^{ij}(T_{\sigma}^{ij}(\mathbf{v}, \mathbf{v}_*)) = \mathbf{v}_*.$$

■

Proposition 4.4. Let $m_i, m_j > 0$ be the masses of particles of species i and j , respectively, $i, j \in \{1, 2, \dots, N\}$, and fix an apse direction $\sigma \in \mathbb{S}^2$. Then,

$$T_{\sigma,1}^{ij}(\mathbf{v}_*, \mathbf{v}) = T_{\sigma,2}^{j,i}(\mathbf{v}, \mathbf{v}_*), \quad (19)$$

$$T_{\sigma,2}^{ij}(\mathbf{v}_*, \mathbf{v}) = T_{\sigma,1}^{j,i}(\mathbf{v}, \mathbf{v}_*). \quad (20)$$

Proof. Let $\mathbf{v}, \mathbf{v}_* \in \mathbb{R}^3$ be arbitrary. Then,

$$\begin{aligned} T_{\sigma,1}^{ij}(\mathbf{v}_*, \mathbf{v}) &= \mathbf{v}_* - [\boldsymbol{\sigma} \cdot (\mathbf{v}_* - \mathbf{v})] \frac{2m_j}{m_i + m_j} \boldsymbol{\sigma} \\ &= \mathbf{v}_* + [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_j}{m_i + m_j} \boldsymbol{\sigma} \\ &= T_{\sigma,2}^{ji}(\mathbf{v}, \mathbf{v}_*). \end{aligned}$$

Similarly,

$$\begin{aligned} T_{\sigma,2}^{ij}(\mathbf{v}_*, \mathbf{v}) &= \mathbf{v} + [\boldsymbol{\sigma} \cdot (\mathbf{v}_* - \mathbf{v})] \frac{2m_i}{m_i + m_j} \boldsymbol{\sigma} \\ &= \mathbf{v} - [\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*)] \frac{2m_i}{m_i + m_j} \boldsymbol{\sigma} \\ &= T_{\sigma,1}^{ji}(\mathbf{v}, \mathbf{v}_*). \end{aligned}$$

■

Proposition 4.5. Let $m_i, m_j > 0$ be the masses of particles of species i and j , respectively, $i, j \in \{1, 2, \dots, N\}$, and fix an apse direction $\boldsymbol{\sigma} \in \mathbb{S}^2$. Suppose that $f : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$ is integrable, that is,

$$\int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |f(\mathbf{v}, \mathbf{v}_*)| \, d\mathbf{v} d\mathbf{v}_* < \infty.$$

Then, $f \circ T_{\sigma}^{ij} : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$ is integrable, and

$$\int_{\mathbb{R}^3} \int_{\mathbb{R}^3} f(\mathbf{v}, \mathbf{v}_*) \, d\mathbf{v} d\mathbf{v}_* = \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} f(T_{\sigma}^{ij}(\mathbf{v}, \mathbf{v}_*)) \, d\mathbf{v} d\mathbf{v}_*. \quad (21)$$

Proof. This follows from the standard change of variable formula and the fact that $T_{\sigma}^{ij}(\mathbf{v}, \mathbf{v}_*)$ is a linear transformation with the property that

$$|\det(T_{\sigma}^{ij})| = 1.$$

■

5 The Boltzmann Collision Operator

Definition 5.1. Let $m_i, m_j > 0$ be the masses of particles of species i and j , respectively, $i, j \in \{1, 2, \dots, N\}$. Suppose that $f_i, f_j : \mathbb{R}^3 \rightarrow [0, \infty)$ are integrable kinetic distributions.

The **general binary i, j collision operator** is defined as

$$Q_{i,j}[f_i, f_j](\mathbf{v}) := \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) d\boldsymbol{\sigma} d\mathbf{v}_*, \quad (22)$$

where

$$\begin{aligned} f'_i &:= f_i(\mathbf{x}, \mathbf{v}', t), & f'_{j,*} &:= f_j(\mathbf{x}, \mathbf{v}'_*, t), \\ f_i &:= f_i(\mathbf{x}, \mathbf{v}, t), & f_{j,*} &:= f_j(\mathbf{x}, \mathbf{v}_*, t); \end{aligned}$$

and the primed coordinates are given precisely by the transformation

$$(\mathbf{v}', \mathbf{v}'_*) = T_{\boldsymbol{\sigma}}^{i,j}(\mathbf{v}, \mathbf{v}_*).$$

$B_{i,j} : \mathbb{R}^3 \times \mathbb{S}^2 \rightarrow [0, \infty)$ is the **collision kernel**, and we say that $B_{i,j}$ has **proper structure** iff

1. for all $i, j \in \{1, 2, \dots, N\}$, for all $\boldsymbol{\sigma} \in \mathbb{S}^2$ and all $\mathbf{w} \in \mathbb{R}^3$,

$$B_{i,j}(\mathbf{w}, \boldsymbol{\sigma}) = b_{i,j}(|\mathbf{w} \cdot \boldsymbol{\sigma}|, |\mathbf{w}|), \quad (23)$$

2. and, for $i \neq j$,

$$\frac{1}{m_i} B_{i,j}(\mathbf{w}, \boldsymbol{\sigma}) = \frac{1}{m_j} B_{j,i}(\mathbf{w}, \boldsymbol{\sigma}), \quad (24)$$

for all $\mathbf{w} \in \mathbb{R}^3$ and all $\boldsymbol{\sigma} \in \mathbb{S}^2$.

Proposition 5.2. Suppose that the collision kernel $B_{i,j} : \mathbb{R}^3 \times \mathbb{S}^2 \rightarrow [0, \infty)$ has proper structure. Let $\mathbf{v}, \mathbf{v}_* \in \mathbb{R}^3$ be arbitrary and set

$$(\mathbf{v}', \mathbf{v}'_*) = T_{\boldsymbol{\sigma}}^{i,j}(\mathbf{v}, \mathbf{v}_*),$$

as usual. Then,

$$B_{i,j}(\mathbf{v}' - \mathbf{v}'_*, \boldsymbol{\sigma}) = B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}). \quad (25)$$

Proof. Recall (13) and (14), respectively,

$$\boldsymbol{\sigma} \cdot (\mathbf{v}' - \mathbf{v}'_*) = -\boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*) \quad \text{and} \quad |\mathbf{v}' - \mathbf{v}'_*| = |\mathbf{v} - \mathbf{v}_*|. \quad (26)$$

Then,

$$B_{i,j}(\mathbf{v}' - \mathbf{v}'_*, \boldsymbol{\sigma}) = b_{i,j}(|(\mathbf{v}' - \mathbf{v}'_*) \cdot \boldsymbol{\sigma}|, |\mathbf{v}' - \mathbf{v}'_*|)$$

$$\begin{aligned}
&= b_{i,j}(| - (\mathbf{v} - \mathbf{v}_*) \cdot \boldsymbol{\sigma}|, |\mathbf{v} - \mathbf{v}_*|) \\
&= b_{i,j}(|(\mathbf{v} - \mathbf{v}_*) \cdot \boldsymbol{\sigma}|, |\mathbf{v} - \mathbf{v}_*|) \\
&= B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}),
\end{aligned}$$

which establishes (25). ■

Definition 5.3. Let $\Omega \subset \mathbb{R}^3$ be a bounded open set. We say that the kinetic distributions $f_i : \Omega \times \mathbb{R}^3 \times [0, T] \rightarrow [0, \infty)$, $i = 1, \dots, N$, satisfy the **multi-species Boltzmann equation** iff, for all $\mathbf{x} \in \Omega$, $\mathbf{v} \in \mathbb{R}^3$, and $t \in (0, T]$,

$$\partial_t f_i(\mathbf{x}, \mathbf{v}, t) + \mathbf{v} \cdot \nabla_{\mathbf{x}} f_i(\mathbf{x}, \mathbf{v}, t) = Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t), \quad (27)$$

where $Q_i[f_i, \mathbf{f}]$, the **cumulative binary collision operator**, is defined as

$$Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) := \sum_{j=1}^N Q_{i,j}[f_i, f_j](\mathbf{x}, \mathbf{v}, t). \quad (28)$$

We often assume that the physical boundary conditions are of the **specular reflection type**, that is, for all $(\mathbf{x}, \mathbf{v}) \in \Gamma_{\text{in}}$, where

$$\Gamma_{\text{in}} := \{(\mathbf{x}, \mathbf{v}) \in \partial\Omega \times \mathbb{R}^3 \mid \mathbf{v} \cdot \mathbf{n}(\mathbf{x}) < 0\},$$

and for all $t \in (0, T]$,

$$f_i(\mathbf{x}, \mathbf{v}, t) = f_i(\mathbf{x}, R(\mathbf{x}, \mathbf{v}), t),$$

where

$$R(\mathbf{x}, \mathbf{v}) := \mathbf{v} - 2[\mathbf{v} \cdot \mathbf{n}(\mathbf{x})] \mathbf{n}(\mathbf{x}),$$

and $\mathbf{n}(\mathbf{x})$ is the outward-pointing unit normal vector on the boundary at $\mathbf{x} \in \partial\Omega$.

Remark 5.4. In the following two results, we will assume that the kinetic distributions are functions only of velocity. This simplification reduces the number of independent variables that must be carried about. However, we lose no generality with these assumptions. More typically, the distributions are functions of space ($\mathbf{x}$) and time (t) as well.

Proposition 5.5. Suppose that $f_i : \mathbb{R}^3 \rightarrow [0, \infty)$ and $\psi : \mathbb{R}^3 \rightarrow \mathbb{R}$ are given and integrable. Then, assuming the collision kernel $B_{i,i}$ has proper structure,

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) \psi(\mathbf{v}) d\mathbf{v}$$

$$= \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{i,*} - f_i f_{i,*}] B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\psi + \psi_* - \psi' - \psi'_*] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v}, \quad (29)$$

where the primed coordinates are determined via

$$(\mathbf{v}', \mathbf{v}'_*) = T_{\boldsymbol{\sigma}}^{i,i}(\mathbf{v}, \mathbf{v}_*),$$

and, for shorthand

$$f_i := f_i(\mathbf{v}), \quad f_{i,*} := f_i(\mathbf{v}_*), \quad f'_i := f_i(\mathbf{v}'), \quad f'_{i,*} := f_i(\mathbf{v}'_*),$$

and

$$\psi := \psi(\mathbf{v}), \quad \psi_* := \psi(\mathbf{v}_*), \quad \psi' := \psi(\mathbf{v}'), \quad \psi'_* := \psi(\mathbf{v}'_*),$$

Proof. Let us set

$$J = \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) \psi(\mathbf{v}) d\mathbf{v}.$$

Then, by definition,

$$\begin{aligned} J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{i,*} - f_i f_{i,*}] B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\ &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\boldsymbol{\sigma},1}^{i,i}(\mathbf{v}, \mathbf{v}_*)) f_i(T_{\boldsymbol{\sigma},2}^{i,i}(\mathbf{v}, \mathbf{v}_*)) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\ &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v}. \end{aligned} \quad (30)$$

The integration variables are dummy variables and may be interchanged. Doing so, we obtain

$$\begin{aligned} J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\boldsymbol{\sigma},1}^{i,i}(\mathbf{v}_*, \mathbf{v})) f_i(T_{\boldsymbol{\sigma},2}^{i,i}(\mathbf{v}_*, \mathbf{v})) B_{i,i}(\mathbf{v}_* - \mathbf{v}, \boldsymbol{\sigma}) \psi(\mathbf{v}_*) d\boldsymbol{\sigma} d\mathbf{v} d\mathbf{v}_* \\ &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}_*) f_i(\mathbf{v}) B_{i,i}(\mathbf{v}_* - \mathbf{v}, \boldsymbol{\sigma}) \psi(\mathbf{v}_*) d\boldsymbol{\sigma} d\mathbf{v} d\mathbf{v}_* \\ &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\boldsymbol{\sigma},1}^{i,i}(\mathbf{v}_*, \mathbf{v})) f_i(T_{\boldsymbol{\sigma},2}^{i,i}(\mathbf{v}_*, \mathbf{v})) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}_*) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\ &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}_*) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v}, \end{aligned}$$

where we have used (23). Observe that, using (19) and (20) (with $i = j$),

$$T_{\boldsymbol{\sigma},1}^{i,i}(\mathbf{v}_*, \mathbf{v}) = T_{\boldsymbol{\sigma},2}^{i,i}(\mathbf{v}, \mathbf{v}_*),$$

$$T_{\boldsymbol{\sigma},2}^{i,i}(\mathbf{v}_*, \mathbf{v}) = T_{\boldsymbol{\sigma},1}^{i,i}(\mathbf{v}, \mathbf{v}_*),$$

so that

$$\begin{aligned}
J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\sigma,2}^{i,i}(\mathbf{v}, \mathbf{v}_*)) f_i(T_{\sigma,1}^{i,i}(\mathbf{v}, \mathbf{v}_*)) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
&\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
&= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}') f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
&\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
&= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}') f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
&\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
&= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{i,*} - f_i f_{i,*}] B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v}. \tag{31}
\end{aligned}$$

Next, using the change of variables formula in Proposition 4.5 and the fact that $I = T_{\sigma}^{i,i} \circ T_{\sigma}^{i,i}$, where I is the identity transformation, we have

$$\begin{aligned}
J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\sigma,1}^{i,i}(\mathbf{v}, \mathbf{v}_*)) f_i(T_{\sigma,2}^{i,i}(\mathbf{v}, \mathbf{v}_*)) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}) d\sigma d\mathbf{v}_* d\mathbf{v} \\
&\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}) d\sigma d\mathbf{v}_* d\mathbf{v} \\
&= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v}' - \mathbf{v}'_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v} \\
&\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}') f_i(\mathbf{v}'_*) B_{i,i}(\mathbf{v}' - \mathbf{v}'_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v}.
\end{aligned}$$

Using (25), we find

$$\begin{aligned}
J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_i(\mathbf{v}_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v} \\
&\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}') f_i(\mathbf{v}'_*) B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v} \\
&= -\frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{i,*} - f_i f_{i,*}] B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v}. \tag{32}
\end{aligned}$$

Finally, with (31) as the starting point, using the change of variables formula in Proposition 4.5, using the fact that $I = T_{\sigma}^{i,i} \circ T_{\sigma}^{i,i}$, and using the identity (25), we deduce that

$$J = -\frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{i,*} - f_i f_{i,*}] B_{i,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v}. \tag{33}$$

The combination of identities (30) – (33) gives the desired result. ■

Proposition 5.6. *For $i \neq j$, if $f_i, f_j : \mathbb{R}^3 \rightarrow [0, \infty)$ and $\psi : \mathbb{R}^3 \rightarrow \mathbb{R}$ are given and integrable, then, assuming the collision kernel B_{ij} has proper structure,*

$$\begin{aligned} & \int_{\mathbb{R}^3} Q_{ij}[f_i, f_j](\mathbf{v}) \psi(\mathbf{v}) d\mathbf{v} \\ &= \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{ij}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\psi - \psi'] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}), \\ &+ \frac{1}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{ji}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\psi_* - \psi'_*] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}), \end{aligned} \quad (34)$$

where, to be precise, for the first integral the primed coordinates are determined via

$$(\mathbf{v}', \mathbf{v}'_*) = T_{\boldsymbol{\sigma}}^{i,j}(\mathbf{v}, \mathbf{v}_*),$$

and, in shorthand,

$$f_i := f_i(\mathbf{v}), \quad f_{j,*} := f_j(\mathbf{v}_*), \quad f'_i := f_i(\mathbf{v}'), \quad f'_{j,*} := f_j(\mathbf{v}'_*), \quad \psi := \psi(\mathbf{v}), \quad \psi' := \psi(\mathbf{v}'),$$

and, for the second integral the primed coordinates are determined via

$$(\mathbf{v}', \mathbf{v}'_*) = T_{\boldsymbol{\sigma}}^{j,i}(\mathbf{v}, \mathbf{v}_*),$$

and, in shorthand,

$$f_j := f_j(\mathbf{v}), \quad f_{i,*} := f_i(\mathbf{v}_*), \quad f'_j := f_j(\mathbf{v}'), \quad f'_{i,*} := f_i(\mathbf{v}'_*), \quad \psi_* := \psi(\mathbf{v}_*), \quad \psi'_* := \psi(\mathbf{v}'_*).$$

Proof. The steps are similar to those in the proof of the previous result. As before, let us set

$$J = \int_{\mathbb{R}^3} Q_{ij}[f_i, f_j](\mathbf{v}) \psi(\mathbf{v}) d\mathbf{v}.$$

Then, by definition,

$$\begin{aligned} J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{ij}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\ &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\boldsymbol{\sigma},1}^{i,j}(\mathbf{v}, \mathbf{v}_*)) f_j(T_{\boldsymbol{\sigma},2}^{i,j}(\mathbf{v}, \mathbf{v}_*)) B_{ij}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \\ &\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_j(\mathbf{v}_*) B_{ij}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) \psi(\mathbf{v}) d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}). \end{aligned} \quad (35)$$

Interchanging the integration variables $\mathbf{v}$ and $\mathbf{v}_*$, we obtain

$$\begin{aligned}
J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\sigma,1}^{ij}(\mathbf{v}_*, \mathbf{v})) f_j(T_{\sigma,2}^{ij}(\mathbf{v}_*, \mathbf{v})) B_{ij}(\mathbf{v}_* - \mathbf{v}, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v} d\mathbf{v}_* \\
&\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}_*) f_j(\mathbf{v}) B_{ij}(\mathbf{v}_* - \mathbf{v}, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v} d\mathbf{v}_* \\
&= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_j(T_{\sigma,2}^{ij}(\mathbf{v}_*, \mathbf{v})) f_i(T_{\sigma,1}^{ij}(\mathbf{v}_*, \mathbf{v})) B_{ij}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
&\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_j(\mathbf{v}) f_i(\mathbf{v}_*) B_{ij}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v},
\end{aligned}$$

where we have used (23). Observe that, using (19) and (20) (with $i \neq j$) and (24), we find that

$$\begin{aligned}
J &= \frac{1}{m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_j(T_{\sigma,1}^{ji}(\mathbf{v}, \mathbf{v}_*)) f_i(T_{\sigma,2}^{ji}(\mathbf{v}, \mathbf{v}_*)) B_{ji}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
&\quad - \frac{1}{m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_j(\mathbf{v}) f_i(\mathbf{v}_*) B_{ji}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \\
&= \frac{1}{m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f_j' f_{i,*}' - f_j f_{i,*}] B_{ji}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\sigma}^{ji}). \quad (36)
\end{aligned}$$

Next, using the change of variables formula in Proposition 4.5 and the fact that $I = T_{\sigma}^{ij} \circ T_{\sigma}^{ij}$, where I is the identity transformation, we have

$$\begin{aligned}
J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(T_{\sigma,1}^{ij}(\mathbf{v}, \mathbf{v}_*)) f_j(T_{\sigma,2}^{ij}(\mathbf{v}, \mathbf{v}_*)) B_{ij}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}) d\sigma d\mathbf{v}_* d\mathbf{v} \\
&\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_j(\mathbf{v}_*) B_{ij}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}) d\sigma d\mathbf{v}_* d\mathbf{v} \\
&= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_j(\mathbf{v}_*) B_{ij}(\mathbf{v}' - \mathbf{v}_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v} \\
&\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}') f_j(\mathbf{v}_*) B_{ij}(\mathbf{v}' - \mathbf{v}_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v}.
\end{aligned}$$

Using (25), we then find

$$\begin{aligned}
J &= \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}) f_j(\mathbf{v}_*) B_{ij}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v} \\
&\quad - \frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} f_i(\mathbf{v}') f_j(\mathbf{v}_*) B_{ij}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v} \\
&= -\frac{1}{m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f_i' f_{j,*}' - f_i f_{j,*}] B_{ij}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}') d\sigma d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\sigma}^{ij}). \quad (37)
\end{aligned}$$

Finally, with (36) as the starting point, using the change of variables formula in Proposi-

tion 4.5, using the fact that $I = T_{\sigma}^{j,i} \circ T_{\sigma}^{j,i}$, and using the identity (25), we deduce that

$$J = -\frac{1}{m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \sigma) \psi(\mathbf{v}_*) d\sigma d\mathbf{v}_* d\mathbf{v} \quad (i \leftrightarrow j). \quad (38)$$

The combination of identities (35) – (38) gives the desired result. $\blacksquare$

6 Collision Invariants

Proposition 6.1. *Suppose that $f_i : \mathbb{R}^3 \rightarrow [0, \infty)$ is given and integrable. Then, assuming the collision kernel $B_{i,i}$ has proper structure, it follows that*

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) d\mathbf{v} = 0, \quad (39)$$

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) \mathbf{v} d\mathbf{v} = \mathbf{0}, \quad (40)$$

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) |\mathbf{v}|^2 d\mathbf{v} = 0. \quad (41)$$

Proof. These results follow from identity (29) in Proposition 5.5 and the transformation invariants (binary collision invariants) in Proposition 4.2 for the case that $i = j$. $\blacksquare$

Definition 6.2. *Suppose that $f_i : \mathbb{R}^3 \rightarrow [0, \infty)$ is given and integrable. Any function $\psi : \mathbb{R}^3 \rightarrow \mathbb{R}$ with the property that*

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) \psi(\mathbf{v}) d\mathbf{v} = 0 \quad (42)$$

*is called a **species i collisional invariant** or a **unary collisional invariant**.*

If the collision kernel $B_{i,i}$ has proper structure, then $\psi \equiv 1$, $\psi(\mathbf{v}) = \mathbf{v}$, and $\psi(\mathbf{v}) = |\mathbf{v}|^2$ are unary collisional invariants.

Proposition 6.3. *For $i \neq j$, if $f_i, f_j : \mathbb{R}^3 \rightarrow [0, \infty)$ are given and integrable, then, assuming the collision kernel $B_{i,j}$ has proper structure, it follows that*

$$\int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) d\mathbf{v} = 0, \quad (43)$$

$$m_i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \mathbf{v} d\mathbf{v} + m_j \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) \mathbf{v} d\mathbf{v} = \mathbf{0}, \quad (44)$$

$$m_i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) |\mathbf{v}|^2 d\mathbf{v} + m_j \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) |\mathbf{v}|^2 d\mathbf{v} = 0. \quad (45)$$

Proof. The identity (43) follows from (34) with $\psi \equiv 1$. Using $\psi(\mathbf{v}) = \mathbf{v}$ in (34), we find that

$$\begin{aligned}
& m_i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \mathbf{v} \, d\mathbf{v} + m_j \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) \mathbf{v} \, d\mathbf{v} \\
&= \frac{m_i}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\mathbf{v} - \mathbf{v}'] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}), \\
&+ \frac{m_i}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\mathbf{v}_* - \mathbf{v}'] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}), \\
&+ \frac{m_j}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\mathbf{v} - \mathbf{v}'] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}), \\
&+ \frac{m_j}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [\mathbf{v}_* - \mathbf{v}'] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}) \\
&= \mathbf{0},
\end{aligned} \tag{46}$$

where we have used the following binary momentum invariants from Proposition 4.2:

$$m_i \mathbf{v} + m_j \mathbf{v}_* = m_i \mathbf{v}' + m_j \mathbf{v}'_* \iff m_i(\mathbf{v} - \mathbf{v}') = m_j(\mathbf{v}'_* - \mathbf{v}_*) \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}), \tag{47}$$

$$m_j \mathbf{v} + m_i \mathbf{v}_* = m_j \mathbf{v}' + m_i \mathbf{v}'_* \iff m_j(\mathbf{v} - \mathbf{v}') = m_i(\mathbf{v}'_* - \mathbf{v}_*) \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}). \tag{48}$$

Using $\psi(\mathbf{v}) = |\mathbf{v}|^2$ in (34) and the appropriate binary kinetic energy invariants from Proposition 4.2, the identity (45) immediately follows. $\blacksquare$

Proposition 6.4. *Suppose that $i, j \in \{1, \dots, N\}$ are arbitrary and $f_i, f_j : \mathbb{R}^3 \rightarrow [0, \infty)$ are given and integrable. Assume that the collision kernel $B_{i,j}$ has proper structure. Recall our definition*

$$Q_i[f_i, \mathbf{f}](\mathbf{v}) := \sum_{j=1}^N Q_{i,j}[f_i, f_j](\mathbf{v}). \tag{49}$$

Then,

$$\int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}) \, d\mathbf{v} = 0, \tag{50}$$

$$\sum_{i=1}^N m_i \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}) \mathbf{v} \, d\mathbf{v} = \mathbf{0}, \tag{51}$$

$$\sum_{i=1}^N m_i \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}) |\mathbf{v}|^2 \, d\mathbf{v} = 0. \tag{52}$$

Proof. We prove only (51). The other two identities are proved similarly. Set

$$J = \sum_{i=1}^N m_i \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}) \mathbf{v} d\mathbf{v}.$$

Then,

$$\begin{aligned} J &= \sum_{i=1}^N m_i \int_{\mathbb{R}^3} \sum_{j=1}^N Q_{i,j}[f_i, f_j](\mathbf{v}) \mathbf{v} d\mathbf{v} \\ &= \sum_{i=1}^N \sum_{j=1}^N m_i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \mathbf{v} d\mathbf{v} \\ &= \sum_{i=2}^N \sum_{j=1}^i \left(m_i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \mathbf{v} d\mathbf{v} + m_j \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) \mathbf{v} d\mathbf{v} \right) \\ &\quad + \sum_{i=1}^N m_i \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i](\mathbf{v}) \mathbf{v} d\mathbf{v} \\ &= \mathbf{0}, \end{aligned}$$

where we have use (40) for the diagonal terms and (44) for the off-diagonal terms. ■

With the next result, we show that all unary collisional invariants have special structure.

Proposition 6.5. *Suppose that $\psi \in C^2(\mathbb{R}^3; \mathbb{R})$ has the property that*

$$\psi(\mathbf{v}) + \psi(\mathbf{v}_*) = \psi(T_{\sigma,1}^{i,i}(\mathbf{v}, \mathbf{v}_*)) + \psi(T_{\sigma,2}^{i,i}(\mathbf{v}, \mathbf{v}_*)), \quad (53)$$

for all $\mathbf{v}, \mathbf{v}_* \in \mathbb{R}^3$ and all $\sigma \in \mathbb{S}^2$. Then, there exist constants $A, C \in \mathbb{R}$ and $\mathbf{B} \in \mathbb{R}^3$, such that

$$\psi(\mathbf{v}) = \frac{m_i A}{2} |\mathbf{v}|^2 + m_i \mathbf{B} \cdot \mathbf{v} + C. \quad (54)$$

In other words, ψ is a velocity-space quadratic function.

Proof. This proof is from [2]. See [4] for another. Define

$$\begin{aligned} \frac{m_i A}{2} &= \frac{1}{6} \Delta \psi(\mathbf{0}), \\ m_i \mathbf{B} &= \nabla \psi(\mathbf{0}), \\ C &= \psi(\mathbf{0}). \end{aligned}$$

Set

$$\phi(\mathbf{v}) = \psi(\mathbf{v}) - \left(\frac{m_i A}{2} |\mathbf{v}|^2 + m_i \mathbf{B} \cdot \mathbf{v} + C \right), \quad \forall \mathbf{v} \in \mathbb{R}^3.$$

Then $\phi \in C^2(\mathbb{R}^3; \mathbb{R})$ is a unary collisional invariant, that is,

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \phi(\mathbf{v}) \, d\mathbf{v} = 0,$$

with the further property that

$$\phi(\mathbf{0}) = 0, \quad \nabla \phi(\mathbf{0}) = \mathbf{0}, \quad \Delta \phi(\mathbf{0}) = 0.$$

Now, suppose that $\mathbf{v}_* \in \mathbb{R}^3 \setminus \{\mathbf{0}\}$ is fixed but, otherwise, arbitrary. Suppose $\boldsymbol{\sigma} \in \mathbb{S}^2$ is perpendicular to $\mathbf{v}_*$, that is,

$$\boldsymbol{\sigma} \cdot \mathbf{v}_* = 0.$$

Set, for $s \in \mathbb{R} \setminus \{0\}$,

$$\mathbf{v} = s\boldsymbol{\sigma}.$$

Then, under the transformation $T_{\boldsymbol{\sigma}}^{i,i}$,

$$\begin{aligned} \mathbf{v}' &:= T_{\boldsymbol{\sigma},1}^{i,i}(\mathbf{v}, \mathbf{v}_*) \\ &= \mathbf{v} - \boldsymbol{\sigma} \cdot (\mathbf{v} - \mathbf{v}_*) \frac{2m_i}{m_i + m_i} \boldsymbol{\sigma} \\ &= \mathbf{v} - s\boldsymbol{\sigma} \\ &= \mathbf{0}. \end{aligned}$$

Furthermore,

$$\mathbf{v}'_* := T_{\boldsymbol{\sigma},2}^{i,i}(\mathbf{v}, \mathbf{v}_*) = \mathbf{v}_* + s\boldsymbol{\sigma}.$$

Thus,

$$\begin{aligned} \phi(s\boldsymbol{\sigma}) + \phi(\mathbf{v}_*) &= \phi(\mathbf{v}) + \phi(\mathbf{v}_*) \\ &= \phi(\mathbf{v}') + \phi(\mathbf{v}'_*) \\ &= \phi(\mathbf{0}) + \phi(\mathbf{v}_* + s\boldsymbol{\sigma}) \\ &= \phi(\mathbf{v}_* + s\boldsymbol{\sigma}). \end{aligned}$$

Equivalently,

$$\phi(s\boldsymbol{\sigma}) = \phi(\mathbf{v}_* + s\boldsymbol{\sigma}) - \phi(\mathbf{v}_*).$$

Using l'Hôpital's rule and the fact that $\phi(\mathbf{0}) = 0$ and $\nabla\phi(\mathbf{0}) = \mathbf{0}$, it follows that

$$\boldsymbol{\sigma} \cdot \nabla\phi(\mathbf{v}_*) = 0.$$

Since $\mathbf{v}_* \in \mathbb{R}^3 \setminus \{\mathbf{0}\}$ and $\boldsymbol{\sigma} \in \mathbb{S}^2$ are perpendicular but otherwise arbitrary, it must be that

$$\phi(\mathbf{v}) = g(|\mathbf{v}|), \quad \forall \mathbf{v} \in \mathbb{R}^3,$$

that is, ϕ is radially symmetric.

Next, assume that $\mathbf{n} \in \mathbb{S}^2$ is arbitrary. Set

$$\mathbf{v} = s\mathbf{n} \quad \text{and} \quad \mathbf{v}_* = r\mathbf{n},$$

where

$$0 \leq s < r.$$

Pick $\boldsymbol{\sigma} \in \mathbb{S}^3$ such that

$$\boldsymbol{\sigma} \cdot \mathbf{n} = \pm \frac{1}{\sqrt{2}}.$$

With this particular choice,

$$\begin{aligned} \mathbf{v}' &= s\mathbf{n} - (s - r) \frac{\pm 1}{\sqrt{2}} \boldsymbol{\sigma}, \\ \mathbf{v}'_* &= r\mathbf{n} + (s - r) \frac{\pm 1}{\sqrt{2}} \boldsymbol{\sigma}, \end{aligned}$$

and

$$\begin{aligned} |\mathbf{v}'|^2 &= \frac{s^2 + r^2}{2}, \\ |\mathbf{v}'_*|^2 &= \frac{s^2 + r^2}{2}. \end{aligned}$$

Since

$$\phi(\mathbf{v}) + \phi(\mathbf{v}_*) = \phi(\mathbf{v}') + \phi(\mathbf{v}'_*),$$

and ϕ is radially symmetric, we have

$$g(s) + g(r) = 2g\left(\sqrt{\frac{s^2 + r^2}{2}}\right).$$

It follows that, upon differentiating with respect to s ,

$$\begin{aligned} g''(s) &= g'' \left(\sqrt{\frac{s^2 + r^2}{2}} \right) \sqrt{\frac{2}{s^2 + r^2}} s + g' \left(\sqrt{\frac{s^2 + r^2}{2}} \right) \sqrt{\frac{2}{s^2 + r^2}} \\ &\quad - g' \left(\sqrt{\frac{s^2 + r^2}{2}} \right) \left(\frac{2}{s^2 + r^2} \right)^{3/2} \frac{s^2}{2}. \end{aligned}$$

Finally, for any $r > 0$,

$$0 = g''(s)|_{s=0} = g' \left(\frac{r}{\sqrt{2}} \right) \frac{\sqrt{2}}{r},$$

which implies that $g' \left(\frac{r}{\sqrt{2}} \right) = 0$, for any $r > 0$. The only possibility is that $g \equiv 0$. ■

7 Information, Entropy, and Boltzmann's H-Theorem

Definition 7.1. Suppose that $f_1, \dots, f_N : \Omega \times \mathbb{R}^3 \times [0, T] \rightarrow [0, \infty)$ are kinetic distributions that satisfy the multi-species Boltzmann equation. The quantity $h : \Omega \times [0, T] \rightarrow \mathbb{R}$, defined via

$$h(\mathbf{x}, t) := \sum_{i=1}^N \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) \ln(f_i(\mathbf{x}, \mathbf{v}, t)) \, d\mathbf{v},$$

is called the **information density** of the multi-species mixture. The quantity

$$H(t) = \int_{\Omega} h(\mathbf{x}, t) \, d\mathbf{x}$$

is called the **total information** of the mixture. The quantity $s : \Omega \times [0, T] \rightarrow \mathbb{R}$ defined via

$$s(\mathbf{x}, t) := - \sum_{i=1}^N \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) \ln(f_i(\mathbf{x}, \mathbf{v}, t)) \, d\mathbf{v} = -h(\mathbf{x}, t),$$

is called the **entropy density** of the multi-species mixture. The quantity

$$S(t) = \int_{\Omega} s(\mathbf{x}, t) \, d\mathbf{x} = -H(t)$$

is called the **total entropy** of the mixture.

We will need the following technical lemma.

Lemma 7.2. *For any $x, y \in (0, \infty)$,*

$$(\ln(x) - \ln(y))(x - y) \geq 0, \quad (55)$$

with equality iff $x = y$.

Proof. Observe that the function $q(x) = x \ln(x)$ is strictly convex on $[0, \infty)$. In fact, for any $x \in (0, \infty)$,

$$q''(x) = \frac{1}{x} > 0.$$

By Taylor's theorem, for any $x, y \in (0, \infty)$, $x \neq y$,

$$x \ln(x) = y \ln(y) + (\ln(y) + 1)(x - y) + \frac{1}{2} \frac{1}{\xi} (x - y)^2 > y \ln(y) + (\ln(y) + 1)(x - y),$$

for some ξ between x and y . The inequality above can be rewritten as

$$x \ln(x) - x \ln(y) > x - y.$$

Reversing the roles of x and y , we have

$$y \ln(y) - y \ln(x) > y - x.$$

Adding the inequalities, we have

$$(x - y)(\ln(x) - \ln(y)) > 0. \quad (56)$$

If $x = y$, then, clearly,

$$(\ln(x) - \ln(y))(x - y) = 0.$$

Conversely, suppose that

$$(\ln(x) - \ln(y))(x - y) = 0.$$

We get a clear contradiction, unless $x = y$, because of the strict inequality (56). ■

Proposition 7.3 (H-Theorem). *Suppose that $f_1, \dots, f_N : \Omega \times \mathbb{R}^3 \times [0, T] \rightarrow [0, \infty)$ are kinetic distributions that satisfy the multi-species Boltzmann equation with specular reflection-type boundary conditions. Assume that the collision kernels B_{ij} have proper structure. Then,*

$$\partial_t h(\mathbf{x}, t) + \nabla_{\mathbf{x}} \cdot \mathbf{J}_h(\mathbf{x}, t) = \sigma_h(\mathbf{x}, t) \leq 0, \quad (57)$$

where $\mathbf{J}_h$ is the **information flux**, given by

$$\mathbf{J}_h(\mathbf{x}, t) := \sum_{i=1}^N \int_{\mathbb{R}^3} \mathbf{v} f_i(\mathbf{x}, \mathbf{v}, t) \ln(f_i(\mathbf{x}, \mathbf{v}, t)) \, d\mathbf{v},$$

and σ_h is the **information loss rate**, defined via

$$\sigma_h(\mathbf{x}, t) := \sum_{i=1}^N \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) \ln(f_i(\mathbf{x}, \mathbf{v}, t)) \, d\mathbf{v}.$$

The total information satisfies

$$d_t H(t) = \int_{\Omega} \sigma_h(\mathbf{x}, t) \, d\mathbf{x} \leq 0, \quad (58)$$

since, for specular reflection-type boundary conditions, it holds that

$$\mathbf{n}(\mathbf{x}) \cdot \mathbf{J}_h(\mathbf{x}, t) = 0, \quad \forall \mathbf{x} \in \partial\Omega, \quad (59)$$

where $\mathbf{n}(\mathbf{x})$ is the outward point unit normal on $\partial\Omega$ at the physical boundary point $\mathbf{x}$. Similarly,

$$\partial_t s(\mathbf{x}, t) + \nabla_{\mathbf{x}} \cdot \mathbf{J}_s(\mathbf{x}, t) = \sigma_s(\mathbf{x}, t) \geq 0, \quad (60)$$

where $\mathbf{J}_s$ is the **entropy flux**, given by

$$\mathbf{J}_s := - \sum_{i=1}^N \int_{\mathbb{R}^3} \mathbf{v} f_i(\mathbf{x}, \mathbf{v}, t) \ln(f_i(\mathbf{x}, \mathbf{v}, t)) \, d\mathbf{v} = -\mathbf{J}_h,$$

and σ_s is the **entropy production rate**, defined via

$$\sigma_s(\mathbf{x}, t) := - \sum_{i=1}^N \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) \ln(f_i(\mathbf{x}, \mathbf{v}, t)) \, d\mathbf{v} = -\sigma_h(\mathbf{x}, t).$$

The total entropy satisfies

$$d_t S(t) = \int_{\Omega} \sigma_s(\mathbf{x}, t) \, d\mathbf{x} \geq 0, \quad (61)$$

since

$$\mathbf{n}(\mathbf{x}) \cdot \mathbf{J}_s(\mathbf{x}, t) = 0, \quad \forall \mathbf{x} \in \partial\Omega. \quad (62)$$

Proof. Let us multiply the i^{th} equation of multi-species Boltzmann system (27) by $\ln(f_i) + 1$, integrate in velocity space, and sum on i . We clearly obtain (57) after using (50).

Next, we want to show that

$$\sigma_h(\mathbf{x}, t) = \sum_{i=1}^N \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) \ln(f_i(\mathbf{x}, \mathbf{v}, t)) \, d\mathbf{v} \leq 0, \quad \forall \mathbf{x} \in \Omega, \quad \forall t \geq 0.$$

Let us write this as the sum of sums, a diagonal sum and an off-diagonal sum:

$$\begin{aligned} \sigma_h &= \sum_{i=1}^N \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) \, d\mathbf{v} \\ &\quad + \sum_{i=2}^N \sum_{j=1}^i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] \ln(f_i) \, d\mathbf{v} + \sum_{j=2}^N \sum_{i=1}^j \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] \ln(f_i) \, d\mathbf{v} \\ &= \sum_{i=1}^N \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) \, d\mathbf{v} \\ &\quad + \sum_{i=2}^N \sum_{j=1}^i \left[\int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] \ln(f_i) \, d\mathbf{v} + \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i] \ln(f_j) \, d\mathbf{v} \right]. \end{aligned} \quad (63)$$

Let us tackle the diagonal terms first. Observe that, using (29),

$$\begin{aligned} &\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) \, d\mathbf{v} \\ &= \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{i,*} - f_i f_{i,*}] B_{i,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_i + L_{i,*} - L'_i - L'_{i,*}] \, d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,i}), \end{aligned} \quad (64)$$

where

$$L_i := \ln(f_i), \quad L_{i,*} := \ln(f_{i,*}), \quad L'_i = \ln(f'_i), \quad L'_{i,*} = \ln(f'_{i,*}).$$

Now, observe that

$$L_i + L_{i,*} - L'_i - L'_{i,*} = \ln(f_i f_{i,*}) - \ln(f'_i f'_{i,*}).$$

Using estimate (55), it follows that

$$[f'_i f'_{i,*} - f_i f_{i,*}] [L_i + L_{i,*} - L'_i - L'_{i,*}] \leq 0,$$

for each $\mathbf{x} \in \Omega$, every $\mathbf{v} \in \mathbb{R}^3$, and every $t \geq 0$. Since $B_{i,i}(\mathbf{w}, \boldsymbol{\sigma}) \geq 0$, for all $\mathbf{w} \in \mathbb{R}^3$ and $\boldsymbol{\sigma} \in \mathbb{S}^2$, it follows that

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) \, d\mathbf{v} \leq 0$$

for each $\mathbf{x} \in \Omega$ and every $t \geq 0$. As a consequence,

$$\sum_{i=1}^N \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) d\mathbf{v} \leq 0.$$

Next we address the off-diagonal sum in (63). Using (34), for $i > j$,

$$\begin{aligned} & \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \ln(f_i(\mathbf{v})) d\mathbf{v} \\ &= \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_i - L'_i] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}) \\ &+ \frac{1}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_{i,*} - L'_{i,*}] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}), \end{aligned}$$

and

$$\begin{aligned} & \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) \ln(f_j(\mathbf{v})) d\mathbf{v} \\ &= \frac{1}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_j - L'_j] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}) \\ &+ \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_{j,*} - L'_{j,*}] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}). \end{aligned}$$

Thus, for $i > j$,

$$\begin{aligned} & \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \ln(f_i(\mathbf{v})) d\mathbf{v} + \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) \ln(f_j(\mathbf{v})) d\mathbf{v} \\ &= \frac{1}{4m_i} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_i f'_{j,*} - f_i f_{j,*}] B_{i,j}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_i - L'_i + L_{j,*} - L'_{j,*}] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{i,j}) \\ &+ \frac{1}{4m_j} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} [f'_j f'_{i,*} - f_j f_{i,*}] B_{j,i}(\mathbf{v} - \mathbf{v}_*, \boldsymbol{\sigma}) [L_{i,*} - L'_{i,*} + L_j - L'_j] d\boldsymbol{\sigma} d\mathbf{v}_* d\mathbf{v} \quad (' \leftrightarrow T_{\boldsymbol{\sigma}}^{j,i}). \end{aligned} \tag{65}$$

Using the logarithmic estimate (55) and the fact that $B_{i,j}, B_{j,i} \geq 0$, we conclude that

$$\int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j](\mathbf{v}) \ln(f_i(\mathbf{v})) d\mathbf{v} + \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i](\mathbf{v}) \ln(f_j(\mathbf{v})) d\mathbf{v} \leq 0.$$

Both terms in (63) are non-positive, and we conclude that $\sigma_h \leq 0$. The estimate (57) is proven.

To prove (58), integrate (57) over Ω , use the Divergence Theorem, the specular boundary condition (59), and the fact that $\sigma_h \leq 0$, point-wise.

The results for the entropy density and the total entropy are simple consequences of the corresponding results for the information density and total information, noting only that $s = -h$ and $S = -H$. ■

Proposition 7.4. *Suppose that $f_1, \dots, f_N : \Omega \times \mathbb{R}^3 \times [0, T] \rightarrow [0, \infty)$ are kinetic distributions. Assume that the collision kernels $B_{i,j}$ have proper structure. Then,*

$$\sigma_h(\mathbf{x}, t) := \sum_{i=1}^N \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) \ln(f_i(\mathbf{x}, \mathbf{v}, t)) \, d\mathbf{v} = 0 \quad (66)$$

iff, for $i = 1, \dots, N$,

$$\ln(f_i(\mathbf{x}, \mathbf{v}, t)) = \frac{m_i A(\mathbf{x}, t)}{2} |\mathbf{v}|^2 + m_i \mathbf{B}(\mathbf{x}, t) \cdot \mathbf{v} + C_i(\mathbf{x}, t), \quad (67)$$

where A and $\mathbf{B}$ are independent of i and $\mathbf{v}$.

Proof. We have seen from the proof of the last proposition that

$$\begin{aligned} \sigma_h &= \sum_{i=1}^N \int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) \, d\mathbf{v} \\ &\quad + \sum_{i=2}^N \sum_{j=1}^i \left[\int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] \ln(f_i) \, d\mathbf{v} + \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i] \ln(f_j) \, d\mathbf{v} \right] \\ &\leq 0. \end{aligned} \quad (68)$$

More specifically,

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) \, d\mathbf{v} \leq 0, \quad i \in \{1, \dots, N\},$$

and

$$\int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] \ln(f_i) \, d\mathbf{v} + \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i] \ln(f_j) \, d\mathbf{v} \leq 0, \quad i, j \in \{1, \dots, N\}, \quad i > j.$$

If (67) holds, then $\sigma_h = 0$, using (39) – (41) ($i = j$) and (43) – (45) ($i > j$). Specifically, for the latter case ($i > j$), observe that

$$\begin{aligned} &\int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] \ln(f_i) \, d\mathbf{v} + \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i] \ln(f_j) \, d\mathbf{v} \\ &= \frac{A(\mathbf{x}, t)}{2} \left(m_i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] |\mathbf{v}|^2 \, d\mathbf{v} + m_j \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i] |\mathbf{v}|^2 \, d\mathbf{v} \right) \\ &\quad + \mathbf{B}(\mathbf{x}, t) \cdot \left(m_i \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] \mathbf{v} \, d\mathbf{v} + m_j \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i] \mathbf{v} \, d\mathbf{v} \right) \end{aligned}$$

$$\begin{aligned}
& + \left(C_i(\mathbf{x}, t) \int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] d\mathbf{v} + C_j(\mathbf{x}, t) \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i] d\mathbf{v} \right) \\
& = 0,
\end{aligned}$$

thanks to (43) – (45).

Conversely, if $\sigma_h(\mathbf{x}, t) = 0$ at a particular point $(\mathbf{x}, t) \in \Omega \times [0, \infty)$, then it must be that

$$\int_{\mathbb{R}^3} Q_{i,i}[f_i, f_i] \ln(f_i) d\mathbf{v} = 0, \quad \forall i \in \{1, \dots, N\}, \quad (69)$$

and

$$\int_{\mathbb{R}^3} Q_{i,j}[f_i, f_j] \ln(f_i) d\mathbf{v} + \int_{\mathbb{R}^3} Q_{j,i}[f_j, f_i] \ln(f_j) d\mathbf{v} = 0, \quad \forall i, j \in \{1, \dots, N\}, \quad i > j. \quad (70)$$

Using Equation (69), Equation (64), and Lemma 7.2, it must be that

$$L_i + L_{i,*} - L'_i - L'_{i,*} = 0 \quad (' \leftrightarrow T_{\sigma}^{i,i}),$$

for all $\mathbf{v}$, $\mathbf{v}_*$, and σ . Invoking Proposition 6.5, it follows that, for $i = 1, \dots, N$, for all $\mathbf{v} \in \mathbb{R}^3$,

$$L_i = \ln(f_i(\mathbf{x}, \mathbf{v}, t)) = \frac{m_i A_i(\mathbf{x}, t)}{2} |\mathbf{v}|^2 + m_i \mathbf{B}_i(\mathbf{x}, t) \cdot \mathbf{v} + C_i(\mathbf{x}, t). \quad (71)$$

Here, A_i and $\mathbf{B}_i$ are independent of $\mathbf{v}$, but we leave open the possibility that A_i and $\mathbf{B}_i$ depend upon i .

Next, using Equation (70), Equation (65), and Lemma 7.2, it must be that

$$L_i - L'_i + L_{j,*} - L'_{j,*} = 0 \quad (' \leftrightarrow T_{\sigma}^{i,j})$$

and

$$L_{i,*} - L'_{i,*} + L_j - L'_j = 0 \quad (' \leftrightarrow T_{\sigma}^{j,i}),$$

for all $\mathbf{v}$, $\mathbf{v}_*$, and σ . We now insert (71)—dropping the $\mathbf{x}$ and t dependencies, since these points are fixed—into the last two equations, to obtain

$$\begin{aligned}
& \frac{m_i A_i}{2} |\mathbf{v}|^2 - \frac{m_i A_i}{2} |\mathbf{v}'|^2 + \frac{m_j A_j}{2} |\mathbf{v}_*|^2 - \frac{m_j A_j}{2} |\mathbf{v}'_*|^2 \\
& + m_i \mathbf{B}_i \cdot \mathbf{v} - m_i \mathbf{B}_i \cdot \mathbf{v}' + m_j \mathbf{B}_j \cdot \mathbf{v}_* - m_j \mathbf{B}_j \cdot \mathbf{v}'_* = 0 \quad (' \leftrightarrow T_{\sigma}^{i,j})
\end{aligned}$$

and

$$\begin{aligned} & \frac{m_i A_i}{2} |\mathbf{v}_\star|^2 - \frac{m_i A_i}{2} |\mathbf{v}'_\star|^2 + \frac{m_j A_j}{2} |\mathbf{v}|^2 - \frac{m_j A_j}{2} |\mathbf{v}'|^2 \\ & + m_i \mathbf{B}_i \cdot \mathbf{v}_\star - m_i \mathbf{B}_i \cdot \mathbf{v}'_\star + m_j \mathbf{B}_j \cdot \mathbf{v} - m_j \mathbf{B}_j \cdot \mathbf{v}' = 0 \quad (' \leftrightarrow T_\sigma^{j,i}), \end{aligned}$$

for all $\mathbf{v}$, $\mathbf{v}_\star$, and σ . Now, we invoke the particle transformation invariances, (11) and (12), to obtain

$$(A_j - A_i) \left(\frac{m_j}{2} |\mathbf{v}_\star|^2 - \frac{m_j}{2} |\mathbf{v}'_\star|^2 \right) + (\mathbf{B}_j - \mathbf{B}_i) \cdot (m_j \mathbf{v}_\star - m_j \mathbf{v}'_\star) = 0 \quad (' \leftrightarrow T_\sigma^{i,j}), \quad (72)$$

for all $\mathbf{v}$, $\mathbf{v}_\star$, and σ , and

$$(A_j - A_i) \left(\frac{m_j}{2} |\mathbf{v}|^2 - \frac{m_j}{2} |\mathbf{v}'|^2 \right) + (\mathbf{B}_j - \mathbf{B}_i) \cdot (m_j \mathbf{v} - m_j \mathbf{v}') = 0 \quad (' \leftrightarrow T_\sigma^{j,i}), \quad (73)$$

for all $\mathbf{v}$, $\mathbf{v}_\star$, and σ .

Let us examine Equation (72). Recall, from the transformation $T_\sigma^{i,j}$,

$$\mathbf{v}'_\star = \mathbf{v}_\star + [\sigma \cdot (\mathbf{v} - \mathbf{v}_\star)] \mu_{i,j} \sigma, \quad \mu_{i,j} := \frac{2m_i}{m_i + m_j}.$$

Let $\mathbf{v}_\star$ be an arbitrary, non-zero vector. Choose

$$\mathbf{v} = \frac{\mu_{i,j} - 2}{\mu_{i,j}} \mathbf{v}_\star, \quad \sigma = \frac{\mathbf{v}_\star}{|\mathbf{v}_\star|}.$$

Then, it follows that $\mathbf{v}'_\star = -\mathbf{v}_\star$. Thus, (72) indicates that

$$(\mathbf{B}_j - \mathbf{B}_i) \cdot \mathbf{v}_\star = 0.$$

Since $\mathbf{v}_\star$ was arbitrary, $\mathbf{B}_j = \mathbf{B}_i$. Since i and j are also arbitrary, it must be that $\mathbf{B}_i = \mathbf{B}$, for all i , for some vector $\mathbf{B}$ that is i -independent. Thus, (72) is equivalent to

$$(A_j - A_i) \left(\frac{m_j}{2} |\mathbf{v}_\star|^2 - \frac{m_j}{2} |\mathbf{v}'_\star|^2 \right) = 0 \quad (' \leftrightarrow T_\sigma^{i,j}),$$

for all $\mathbf{v}$, $\mathbf{v}_\star$, and σ . Again, let $\mathbf{v}_\star$ be an arbitrary, non-zero vector. Choose

$$\mathbf{v} = \frac{\mu_{i,j} - 2}{\mu_{i,j}} \mathbf{v}_\star, \quad \sigma = \frac{\mathbf{v}_\star}{|\mathbf{v}_\star|}.$$

Then, it follows that $\mathbf{v}'_\star = \mathbf{0}$. Thus,

$$(A_j - A_i)|\mathbf{v}_\star|^2 = 0.$$

As above, it must be that $A_i = A$, for all i , where A is i -independent. ■

8 Moments and Moment Equations

In this section, we develop the macroscopic moment equations from the Boltzmann equations.

Definition 8.1. Suppose that $f_i : \Omega \times \mathbb{R}^3 \times [0, \infty) \rightarrow [0, \infty)$, $i = 1, \dots, N$ are kinetic distributions with the property that

$$\int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2, \quad (74)$$

for each $\mathbf{x} \in \Omega$ and every $t \geq 0$. Assume that m_i , the **mass of the i^{th} particle**, is a positive real number. The quantity

$$n_i(\mathbf{x}, t) := \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) d\mathbf{v} \quad (75)$$

is called the **number density of the i^{th} species**, while

$$\rho_i(\mathbf{x}, t) := m_i n_i(\mathbf{x}, t) \quad (76)$$

is called the **mass density of the i^{th} species**. The quantity

$$\mathbf{u}_i(\mathbf{x}, t) := \frac{1}{n_i(\mathbf{x}, t)} \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) \mathbf{v} d\mathbf{v} \quad (77)$$

is called the **bulk velocity of the i^{th} species**. The **total mass density of the mixture** is defined as

$$\rho(\mathbf{x}, t) := \sum_{i=1}^N \rho_i(\mathbf{x}, t). \quad (78)$$

The quantity

$$\mathbf{q}(\mathbf{x}, t) := \sum_{i=1}^N \rho_i(\mathbf{x}) \mathbf{u}_i(\mathbf{x}, t) \quad (79)$$

is called the **total momentum density of the mixture**. The **bulk mass-averaged velocity** is defined as

$$\mathbf{u}(\mathbf{x}, t) := \frac{\mathbf{q}(\mathbf{x}, t)}{\rho(\mathbf{x}, t)}. \quad (80)$$

The **kinetic energy density of the i^{th} species** is defined as

$$\omega_i(\mathbf{x}, t) := \frac{m_i}{2} \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^2 d\mathbf{v}. \quad (81)$$

The **total kinetic energy density** is defined as

$$\omega(\mathbf{x}, t) := \sum_{i=1}^N \omega_i(\mathbf{x}, t). \quad (82)$$

The **internal energy density of the i^{th} species** is defined via

$$e_i(\mathbf{x}, t) := \frac{m_i}{2} \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v} - \mathbf{u}_i(\mathbf{x}, t)|^2 d\mathbf{v}. \quad (83)$$

The **total internal energy density** is defined as

$$e(\mathbf{x}, t) := \sum_{i=1}^N e_i(\mathbf{x}, t). \quad (84)$$

Finally, the **temperature of the i^{th} species** is defined via

$$\theta_i(\mathbf{x}, t) := \frac{2 e_i(\mathbf{x}, t)}{3 n_i(\mathbf{x}, t)}, \quad (85)$$

so that, after some simple manipulation,

$$\theta_i(\mathbf{x}, t) = \frac{m_i}{3 n_i(\mathbf{x}, t)} \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v} - \mathbf{u}_i(\mathbf{x}, t)|^2 d\mathbf{v}. \quad (86)$$

Remark 8.2. Clearly, θ_i , the temperature of the i^{th} species is non-negative, provided f_i is non-negative.

Remark 8.3. The quantities ρ_i , $\mathbf{u}_i$, and θ_i are sometimes referred to, collectively, as the **first three moments of f_i** . Other authors use the collection ρ_i , $\rho_i \mathbf{u}_i$, and ω_i to stand for first three moments of f_i . Observe that the collections $(\rho_i, \mathbf{u}_i, \theta_i)$ and $(\rho_i, \rho_i \mathbf{u}_i, \omega_i)$ are essentially equivalent. One collection of variables can be derived from the other and vice versa.

Let us define a few more bulk/macroscopic quantities.

Definition 8.4. Suppose that $f_i : \Omega \times \mathbb{R}^3 \times [0, \infty) \rightarrow [0, \infty)$, $i = 1, \dots, N$ are kinetic distributions with the properties

$$\int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2, 3, \quad (87)$$

for each $\mathbf{x} \in \Omega$ and every $t \geq 0$. The **energy flow of the i^{th} species** is defined as

$$\mathbf{r}_i(\mathbf{x}, t) := \frac{m_i}{2} \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) \mathbf{v} |\mathbf{v}|^2 d\mathbf{v}. \quad (88)$$

The i^{th} **momentum flow tensor** is defined as

$$\mathbf{M}_i(\mathbf{x}, t) := m_i \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) \mathbf{v} \otimes \mathbf{v} d\mathbf{v}. \quad (89)$$

The i^{th} **deviation velocity** is defined as

$$\mathbf{c}_i(\mathbf{x}, \mathbf{v}, t) := \mathbf{v} - \mathbf{u}_i(\mathbf{x}, t). \quad (90)$$

The **pressure tensor of the i^{th} species** is defined as

$$\mathbf{P}_i(\mathbf{x}, t) := m_i \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) \mathbf{c}_i(\mathbf{x}, \mathbf{v}, t) \otimes \mathbf{c}_i(\mathbf{x}, \mathbf{v}, t) d\mathbf{v}. \quad (91)$$

Define the i^{th} **heat flux** via

$$\boldsymbol{\chi}_i(\mathbf{x}, t) := \frac{m_i}{2} \int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) \mathbf{c}_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{c}_i(\mathbf{x}, \mathbf{v}, t)|^2 d\mathbf{v}. \quad (92)$$

Next, we list some elementary identities that relate our defined bulk/macroscopic quantities.

Proposition 8.5. *Suppose that $f_i : \Omega \times \mathbb{R}^3 \times [0, \infty) \rightarrow [0, \infty)$, $i = 1, \dots, N$, are kinetic distributions with the properties*

$$\int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2, 3, \quad (93)$$

for each $\mathbf{x} \in \Omega$ and every $t \geq 0$. Then, we have the following identities:

$$\omega_i(\mathbf{x}, t) = \frac{\rho_i(\mathbf{x}, t)}{2} |\mathbf{u}_i(\mathbf{x}, t)|^2 + e_i(\mathbf{x}, t); \quad (94)$$

$$\mathbf{M}_i(\mathbf{x}, t) = \rho_i(\mathbf{x}, t) \mathbf{u}_i(\mathbf{x}, t) \otimes \mathbf{u}_i(\mathbf{x}, t) + \mathbf{P}_i(\mathbf{x}, t); \quad (95)$$

and

$$\mathbf{r}_i(\mathbf{x}, t) = \omega_i(\mathbf{x}, t) \mathbf{u}_i(\mathbf{x}, t) + \boldsymbol{\chi}_i(\mathbf{x}, t) + \mathbf{P}_i(\mathbf{x}, t) \mathbf{u}_i(\mathbf{x}, t). \quad (96)$$

Proof. The proofs are straightforward and are skipped for brevity. ■

Finally, we can state the basic moment equations for the multi-species Boltzmann system.

Proposition 8.6. *Suppose that $f_i : \Omega \times \mathbb{R}^3 \times [0, \infty) \rightarrow [0, \infty)$, $i = 1, \dots, N$, are kinetic distributions that satisfy the multi-species Boltzmann system,*

$$\partial_t f_i(\mathbf{x}, \mathbf{v}, t) + \mathbf{v} \cdot \nabla_{\mathbf{x}} f_i(\mathbf{x}, \mathbf{v}, t) = Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t), \quad (97)$$

where $Q_i[f_i, \mathbf{f}]$ is the cumulative binary collision operator

$$Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) := \sum_{j=1}^N Q_{ij}[f_i, f_j](\mathbf{x}, \mathbf{v}, t). \quad (98)$$

Assume that

$$\int_{\mathbb{R}^3} f_i(\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2, 3, \quad (99)$$

for each $\mathbf{x} \in \Omega$ and every $t \geq 0$. Suppose that the collision kernels, B_{ij} have proper structure. Then, for all $\mathbf{x} \in \Omega$ and $t \geq 0$, and for each $i = 1, \dots, N$,

$$\partial_t \rho_i + \nabla_{\mathbf{x}} \cdot (\rho_i \mathbf{u}_i) = 0, \quad (100)$$

$$\partial_t (\rho_i \mathbf{u}_i) + \nabla_{\mathbf{x}} \cdot (\rho_i \mathbf{u}_i \otimes \mathbf{u}_i + \mathbf{P}_i) = m_i \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) \mathbf{v} d\mathbf{v}, \quad (101)$$

and

$$\begin{aligned} \partial_t \left(\frac{\rho_i}{2} |\mathbf{u}_i|^2 + e_i \right) + \nabla_{\mathbf{x}} \cdot \left[\left(\frac{\rho_i}{2} |\mathbf{u}_i|^2 + e_i \right) \mathbf{u}_i \right] + \nabla_{\mathbf{x}} \cdot (\mathbf{x}_i + \mathbf{P}_i \mathbf{u}_i) \\ = \frac{m_i}{2} \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^2 d\mathbf{v}. \end{aligned} \quad (102)$$

Consequently, using the collisional invariants,

$$\partial_t \rho + \nabla_{\mathbf{x}} \cdot (\rho \mathbf{u}) = 0, \quad (103)$$

$$\partial_t (\rho \mathbf{u}) + \nabla_{\mathbf{x}} \cdot \sum_{i=1}^N (\rho_i \mathbf{u}_i \otimes \mathbf{u}_i + \mathbf{P}_i) = 0, \quad (104)$$

and

$$\partial_t \sum_{i=1}^N \left(\frac{\rho_i}{2} |\mathbf{u}_i|^2 + e_i \right) + \nabla_{\mathbf{x}} \cdot \sum_{i=1}^N \left[\left(\frac{\rho_i}{2} |\mathbf{u}_i|^2 + e_i \right) \mathbf{u}_i + \mathbf{x}_i + \mathbf{P}_i \mathbf{u}_i \right] = 0. \quad (105)$$

Proof. To prove the first three identities, multiply (97) by, respectively, m_i , $m_i \mathbf{v}$, and $\frac{m_i}{2} |\mathbf{v}|^2$ and integrate over velocity space. The last three are obtained by summing the first three on i and using the usual collision invariants (50) – (52).

Here we will prove only identities (101) and (102) in detail. First, multiply (97) by $m_i \mathbf{v}$ and integrate over velocity space to obtain

$$\begin{aligned} m_i \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) \mathbf{v} d\mathbf{v} &= \partial_t (\rho_i \mathbf{u}_i) + m_i \int_{\mathbb{R}^3} \mathbf{v} \mathbf{v} \cdot \nabla_{\mathbf{x}} f_i d\mathbf{v} \\ &= \partial_t (\rho_i \mathbf{u}_i) + \nabla_{\mathbf{x}} \cdot \mathbf{M}_i \\ &= \partial_t (\rho_i \mathbf{u}_i) + \nabla_{\mathbf{x}} \cdot (\rho_i \mathbf{u}_i \otimes \mathbf{u}_i + \mathbf{P}_i), \end{aligned}$$

where we have used identity (95). Thus, we have identity (101).

Next, multiply (97) by $\frac{m_i}{2} |\mathbf{v}|^2$ and integrate over velocity space to obtain

$$\begin{aligned} \frac{m_i}{2} \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^2 d\mathbf{v} &= \partial_t \omega_i + \nabla_{\mathbf{x}} \cdot \mathbf{r}_i \\ &= \partial_t \omega_i + \nabla_{\mathbf{x}} \cdot (\omega_i \mathbf{u}_i + \boldsymbol{\chi}_i + \mathbf{P}_i \mathbf{u}_i) \\ &= \partial_t \left(\frac{\rho_i}{2} |\mathbf{u}_i|^2 + e_i \right) + \nabla_{\mathbf{x}} \cdot \left[\left(\frac{\rho_i}{2} |\mathbf{u}_i|^2 + e_i \right) \mathbf{u}_i \right] \\ &\quad + \nabla_{\mathbf{x}} \cdot (\boldsymbol{\chi}_i + \mathbf{P}_i \mathbf{u}_i), \end{aligned}$$

using (96) and (94), which gives the identity (102). ■

9 The Maxwellian and Maximal Entropy States

In this section, we will do our calculations in a physical space of generic dimension $d \in \mathbb{N}$, to maintain some generality.

Definition 9.1. Suppose that $\Omega \subset \mathbb{R}^d$ is an open, bounded set and $f : \Omega \times \mathbb{R}^d \times [0, \infty) \rightarrow [0, \infty)$ is a kinetic distribution of particles of mass $m > 0$, with the property

$$\int_{\mathbb{R}^d} f(\mathbf{x}, \mathbf{v}, t) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2, \quad (106)$$

for each $\mathbf{x} \in \Omega$ and every $t \geq 0$. The **Maxwellian distribution of f** is defined as

$$M_f(\mathbf{x}, \mathbf{v}, t) := M[f](\mathbf{x}, \mathbf{v}, t) := n \left(\frac{m}{2\pi\theta} \right)^{d/2} \exp \left(-\frac{m|\mathbf{v} - \mathbf{u}|^2}{2\theta} \right), \quad (107)$$

using the first three integral moments of f ,

$$n(\mathbf{x}, t) := \int_{\mathbb{R}^d} f d\mathbf{v}, \quad \mathbf{u}(\mathbf{x}, t) := \frac{1}{n} \int_{\mathbb{R}^d} f \mathbf{v} d\mathbf{v}, \quad \theta(\mathbf{x}, t) := \frac{m}{d n} \int_{\mathbb{R}^d} f |\mathbf{v} - \mathbf{u}|^2 d\mathbf{v}. \quad (108)$$

For convenience, we define the bracket notation, $\langle \cdot \rangle : L^1(\mathbb{R}^d; \mathbb{R}^p) \rightarrow \mathbb{R}^p$, by

$$\langle \mathbf{g} \rangle = \int_{\mathbb{R}^d} \mathbf{g}(\mathbf{v}) \, d\mathbf{v}.$$

Therefore, we have the usual definitions

$$\begin{bmatrix} \rho(\mathbf{x}, t) \\ \rho(\mathbf{x}, t) \mathbf{u}(\mathbf{x}, t) \\ \omega(\mathbf{x}, t) \end{bmatrix} = \left\langle f(\mathbf{x}, \mathbf{v}, t) \begin{bmatrix} m \\ m\mathbf{v} \\ \frac{m}{2} |\mathbf{v}|^2 \end{bmatrix} \right\rangle,$$

where $m > 0$ is the mass of the particles with kinetic distribution f .

For convenience, let us define

$$\boldsymbol{\rho}(\mathbf{x}, t) := [\rho, \rho \mathbf{u}^\top, \omega]^\top : \mathbb{R}^d \times \mathbb{R}^+ \rightarrow \mathbb{R}^{d+2}; \quad (109)$$

$$\mathbf{m}(\mathbf{v}) := \left[m, m\mathbf{v}^\top, \frac{m}{2} |\mathbf{v}|^2 \right]^\top : \mathbb{R}^d \rightarrow \mathbb{R}^{d+2}; \quad (110)$$

and

$$\boldsymbol{\alpha}(\mathbf{x}, t) := \frac{1}{m} \left[\ln \left(\frac{n}{(2\pi\theta/m)^{d/2}} \right) - \frac{m|\mathbf{u}|^2}{2\theta}, \frac{m\mathbf{u}^\top}{\theta}, -\frac{m}{\theta} \right]^\top : \mathbb{R}^d \times \mathbb{R}^+ \rightarrow \mathbb{R}^{d+2}. \quad (111)$$

Then, succinctly,

$$\boldsymbol{\rho} = \langle f \mathbf{m} \rangle, \quad \text{and} \quad M_f = \exp(\boldsymbol{\alpha} \cdot \mathbf{m}(\mathbf{v})). \quad (112)$$

Indeed,

$$\begin{aligned} \exp(\boldsymbol{\alpha} \cdot \mathbf{m}) &= \exp \left(\ln \left(\frac{n}{(2\pi\theta/m)^{d/2}} \right) - \frac{m|\mathbf{u}|^2}{2\theta} \right) \exp \left(\frac{m\mathbf{u} \cdot \mathbf{v}}{\theta} \right) \exp \left(-\frac{m}{2\theta} |\mathbf{v}|^2 \right) \\ &= n \left(\frac{m}{2\pi\theta} \right)^{d/2} \exp \left(-\frac{m|\mathbf{u}|^2}{2\theta} \right) \exp \left(\frac{m\mathbf{u} \cdot \mathbf{v}}{\theta} \right) \exp \left(-\frac{m}{2\theta} |\mathbf{v}|^2 \right) \\ &= n \left(\frac{m}{2\pi\theta} \right)^{d/2} \exp \left(-\frac{m}{2\theta} |\mathbf{u} - \mathbf{v}|^2 \right) \\ &= M_f. \end{aligned}$$

The next result indicates that the first three moments of f and its associated Maxwellian are exactly the same.

Lemma 9.2. *Suppose that $f : \mathbb{R}^d \rightarrow [0, \infty)$ is a kinetic distribution of particles of mass*

$m > 0$, with the property

$$\int_{\mathbb{R}^d} f(\mathbf{v}) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2. \quad (113)$$

The following equality holds:

$$\langle (M_f - f) \mathbf{m} \rangle = \left\langle \begin{bmatrix} m \\ m\mathbf{v} \\ \frac{m}{2} |\mathbf{v}|^2 \end{bmatrix} (M_f - f) \right\rangle = \mathbf{0}. \quad (114)$$

Proof. 1. First we show that

$$\int_{\mathbb{R}^d} M_f d\mathbf{v} = \int_{\mathbb{R}^d} f d\mathbf{v}. \quad (115)$$

By definition,

$$\begin{aligned} \int_{\mathbb{R}^d} M_f d\mathbf{v} &= \int_{\mathbb{R}^d} \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \exp\left(-\frac{m|\mathbf{v} - \mathbf{u}|^2}{2\theta}\right) d\mathbf{v} \\ &= \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \int_{\mathbb{R}^d} \exp\left(-\left|\frac{\mathbf{v} - \mathbf{u}}{(2\theta/m)^{\frac{1}{2}}}\right|^2\right) d\mathbf{v}. \end{aligned} \quad (116)$$

Consider the substitution

$$\mathbf{s} = \frac{\mathbf{v} - \mathbf{u}}{(2\theta/m)^{\frac{1}{2}}} \implies d\mathbf{s} = \left(\frac{1}{(2\theta/m)^{\frac{1}{2}}}\right)^d d\mathbf{v} \implies d\mathbf{v} = (2\theta/m)^{\frac{d}{2}} d\mathbf{s}. \quad (117)$$

The integral becomes

$$\begin{aligned} \int_{\mathbb{R}^d} M_f d\mathbf{v} &= \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \int_{\mathbb{R}^d} \exp(-|\mathbf{s}|^2) (2\theta/m)^{\frac{d}{2}} d\mathbf{s} \\ &= \frac{n(2\theta/m)^{\frac{d}{2}}}{\pi^{\frac{d}{2}} (2\theta/m)^{\frac{d}{2}}} \int_{\mathbb{R}^d} e^{-|\mathbf{s}|^2} d\mathbf{s} \\ &= \frac{n}{\pi^{\frac{d}{2}}} \cdot \pi^{\frac{d}{2}} \\ &= \int_{\mathbb{R}^d} f d\mathbf{v}, \end{aligned} \quad (118)$$

as desired. Note that we have used the definition $n = \int_{\mathbb{R}^d} f d\mathbf{v}$, and the fact that in d dimensions, the integral $\int_{\mathbb{R}^d} e^{-|\mathbf{x}|^2} d\mathbf{x} = \pi^{\frac{d}{2}}$. We will use this frequently.

2. Next we show that

$$\int_{\mathbb{R}^d} \mathbf{v} M_f d\mathbf{v} = \int_{\mathbb{R}^d} \mathbf{v} f d\mathbf{v}. \quad (119)$$

Following a similar line of work, we have

$$\begin{aligned} \int_{\mathbb{R}^d} \mathbf{v} M_f d\mathbf{v} &= \int_{\mathbb{R}^d} \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \mathbf{v} \exp\left(-\frac{|\mathbf{v}-\mathbf{u}|^2}{2\theta/m}\right) d\mathbf{v} \\ &= \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \int_{\mathbb{R}^d} \mathbf{v} \exp\left(-\left|\frac{\mathbf{v}-\mathbf{u}}{(2\theta/m)^{\frac{1}{2}}}\right|^2\right) d\mathbf{v}. \end{aligned} \quad (120)$$

Using the substitution in (117), the integral becomes

$$\begin{aligned} \int_{\mathbb{R}^d} \mathbf{v} M_f d\mathbf{v} &= \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \int_{\mathbb{R}^d} \left((2\theta/m)^{\frac{1}{2}} \mathbf{s} + \mathbf{u}\right) \exp(-|\mathbf{s}|^2) \cdot (2\theta/m)^{\frac{d}{2}} d\mathbf{s} \\ &= \frac{n(2\theta/m)^{\frac{d}{2}}}{\pi^{\frac{d}{2}}(2\theta/m)^{\frac{d}{2}}} \left[(2\theta/m)^{\frac{1}{2}} \int_{\mathbb{R}^d} \mathbf{s} e^{-|\mathbf{s}|^2} d\mathbf{s} + \mathbf{u} \int_{\mathbb{R}^d} e^{-|\mathbf{s}|^2} d\mathbf{s}\right] \\ &= \frac{n}{\pi^{\frac{d}{2}}} \left[(2\theta/m)^{\frac{1}{2}} \int_{\mathbb{R}^d} \mathbf{s} e^{-|\mathbf{s}|^2} d\mathbf{s} + \mathbf{u} \int_{\mathbb{R}^d} e^{-|\mathbf{s}|^2} d\mathbf{s}\right]. \end{aligned} \quad (121)$$

Let us deal with the first term in (121). One can easily show that

$$\int_{-\infty}^{\infty} x e^{-x^2} dx = 0 \implies \int_{\mathbb{R}^d} \mathbf{s} e^{-|\mathbf{s}|^2} d\mathbf{s} = \mathbf{0}.$$

Therefore, we have

$$\int_{\mathbb{R}^d} \mathbf{v} M_f d\mathbf{v} = \frac{n}{\pi^{\frac{d}{2}}} \cdot \mathbf{u} \int_{\mathbb{R}^d} e^{-|\mathbf{s}|^2} d\mathbf{s} = \frac{n\mathbf{u}}{\pi^{\frac{d}{2}}} \cdot \pi^{\frac{d}{2}} = n\mathbf{u} = \int_{\mathbb{R}^d} \mathbf{v} f d\mathbf{v}.$$

3. Finally, we show that

$$\int \frac{|\mathbf{v}|^2}{2} M_f d\mathbf{v} = \int \frac{|\mathbf{v}|^2}{2} f d\mathbf{v}. \quad (122)$$

First, observe that

$$\int_{\mathbb{R}^d} |\mathbf{s}|^2 e^{-|\mathbf{s}|^2} d\mathbf{s} = \frac{d}{2} \pi^{\frac{d}{2}}. \quad (123)$$

(To see how to prove such results, see Appendix A). Now, using the substitution in (117), and the same integration techniques on the LHS of (122),

$$\int \frac{|\mathbf{v}|^2}{2} M_f d\mathbf{v} = \frac{1}{2} \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \int |\mathbf{v}|^2 e^{-\frac{|\mathbf{v}-\mathbf{u}|^2}{2\theta/m}} d\mathbf{v}$$

$$\begin{aligned}
&= \frac{1}{2} \frac{n}{(2\pi\theta/m)^{\frac{d}{2}}} \int |\mathbf{u} + (\mathbf{v} - \mathbf{u})|^2 e^{-\frac{|\mathbf{v}-\mathbf{u}|^2}{2\theta/m}} d\mathbf{v} \\
&= \frac{n}{2(2\pi\theta/m)^{\frac{d}{2}}} \int (-|\mathbf{u}|^2 + 2\mathbf{u}^\top \mathbf{v} + |\mathbf{v} - \mathbf{u}|^2) e^{-\frac{|\mathbf{v}-\mathbf{u}|^2}{2\theta/m}} d\mathbf{v} \\
&= \frac{-n|\mathbf{u}|^2}{2(2\pi\theta/m)^{\frac{d}{2}}} (2\theta/m)^{\frac{d}{2}} \int_{\mathbb{R}^d} e^{-|\mathbf{s}|^2} d\mathbf{s} \\
&\quad + \frac{2n(2\theta/m)^{\frac{d}{2}}}{2(2\pi\theta/m)^{\frac{d}{2}}} \mathbf{u}^\top \int_{\mathbb{R}^d} [\mathbf{u} + (2\theta/m)^{\frac{1}{2}} \mathbf{s}] e^{-|\mathbf{s}|^2} d\mathbf{s} \\
&\quad + \frac{n(2\theta/m)^{\frac{d}{2}+1}}{2(2\pi\theta/m)^{\frac{d}{2}}} \int_{\mathbb{R}^d} |\mathbf{s}|^2 e^{-|\mathbf{s}|^2} d\mathbf{s} \\
&= \frac{-n|\mathbf{u}|^2}{2\pi^{\frac{d}{2}}} \pi^{\frac{d}{2}} + \frac{n|\mathbf{u}|^2}{\pi^{\frac{d}{2}}} \pi^{\frac{d}{2}} + 0 + \frac{2n\theta/m}{2\pi^{\frac{d}{2}}} \frac{d}{2} \pi^{\frac{d}{2}} \\
&= \frac{1}{2} n|\mathbf{u}|^2 + \frac{d}{2} n\theta/m \\
&= \int_{\mathbb{R}^d} \frac{|\mathbf{v}|^2}{2} f d\mathbf{v}, \tag{124}
\end{aligned}$$

where the final line follows from identity (94) (with d and 3 interchanged).

The proof is complete. ■

We need the following simple result.

Lemma 9.3. *Suppose that $f : \mathbb{R}^d \rightarrow [0, \infty)$ is a kinetic distribution of particles of mass $m > 0$, with the property*

$$\int_{\mathbb{R}^d} f(\mathbf{v}) |\mathbf{v}|^k d\mathbf{v} < \infty, \quad k = 0, 1, 2. \tag{125}$$

The function $\ln(M_f)$ is a Maxwellian invariant, namely,

$$\int_{\mathbb{R}^d} \ln(M_f(\mathbf{v})) (M_f(\mathbf{v}) - f(\mathbf{v})) d\mathbf{v} = \langle \ln(M_f)(M_f - f) \rangle = 0.$$

Proof. From the last result, we already know that, for any $C_0, C_2 \in \mathbb{R}$, and any $\mathbf{C}_1 \in \mathbb{R}^d$,

$$\langle (C_0 + \mathbf{C}_1 \cdot \mathbf{v} + C_2 |\mathbf{v}|^2)(M_f - f) \rangle = 0.$$

Clearly,

$$\ln(M_f) = \ln(\exp(\boldsymbol{\alpha} \cdot \mathbf{m})) = \boldsymbol{\alpha} \cdot \mathbf{m},$$

is a linear combination of 1, $\mathbf{v}$, and $|\mathbf{v}|^2$. It follows that $\ln(M_f)$ is also a Maxwellian invariant, that is, $\langle \ln(M_f)(M_f - f) \rangle = 0$. ■

The Maxwellian is the least information, or information minimizing (entropy maximizing), state. Consider the following result.

Theorem 9.4. *Suppose that*

$$f_\star : \mathbb{R}^d \rightarrow [0, \infty)$$

is a given kinetic distribution. Let

$$\boldsymbol{\rho}_\star = [\rho_\star, \rho_\star \mathbf{u}_\star^\top, \omega_\star]^\top,$$

be the vector comprised of the first three moments of $f_\star$, that is,

$$\boldsymbol{\rho}_\star = \begin{bmatrix} \rho_\star \\ \rho_\star \mathbf{u}_\star \\ \omega_\star \end{bmatrix} = \left\langle f_\star(\mathbf{v}) \begin{bmatrix} m \\ m\mathbf{v} \\ \frac{m}{2}|\mathbf{v}|^2 \end{bmatrix} \right\rangle.$$

The temperature, $\theta_\star$, is defined via

$$\omega_\star = \frac{\rho_\star}{2}|\mathbf{u}_\star|^2 + e_\star = \frac{\rho_\star}{2}|\mathbf{u}_\star|^2 + \frac{d}{2}n_\star\theta_\star.$$

The Maxwellian, as usual, is defined using the macroscopic densities as

$$M_\star(\mathbf{x}, \mathbf{v}) := n_\star \left(\frac{m}{2\pi\theta_\star} \right)^{d/2} \exp \left(-\frac{m|\mathbf{v} - \mathbf{u}_\star|^2}{2\theta_\star} \right).$$

Consider the admissible class of distribution functions, defined as

$$\mathcal{A}_\star := \{g : \mathbb{R}^d \rightarrow \mathbb{R} \mid g \geq 0 \text{ and } \langle m\mathbf{g} \rangle = \boldsymbol{\rho}_\star\},$$

where, recall,

$$\mathbf{m}(\mathbf{v}) := \left[m, m\mathbf{v}^\top, \frac{m}{2}|\mathbf{v}|^2 \right]^\top : \mathbb{R}^d \rightarrow \mathbb{R}^{d+2}.$$

Then, $M_\star, f_\star \in \mathcal{A}_\star$. In other words, $\mathcal{A}_\star$ is non-empty. For every $g \in \mathcal{A}_\star$,

$$h[M_\star] \leq h[g],$$

where $h[g]$ is the information content of g , defined as

$$h[g] := \int_{\mathbb{R}^d} g(\mathbf{v}) \ln(g(\mathbf{v})) \, d\mathbf{v}.$$

In other words, the Maxwellian is the minimum information distribution out of the set of admissible distributions that gives rise to the macroscopic density ρ_* .

Proof. Let us first show that $M_* \in \mathcal{A}_*$. Clearly M_* is non-negative. The fact that $\langle \mathbf{m} M_* \rangle = \rho_*$ follows from Lemma 9.2. Thus $M_* \in \mathcal{A}_*$.

Next, we prove the inequality. Let $g \in \mathcal{A}_*$ be arbitrary. We may write $M_* = M_g$, since g gives rise to the same macroscopic fields as f_* . Define $\hat{h}(s) := s \ln(s)$. This function $\hat{h}$ is convex on $(0, \infty)$, as is easy to show. Its first two derivatives are

$$\hat{h}'(s) = \ln(s) + 1, \quad \hat{h}''(s) = \frac{1}{s} > 0, \quad \forall s > 0.$$

Thus,

$$\hat{h}(s) \geq \hat{h}(t) + \hat{h}'(t)(s - t), \quad \forall s, t \in (0, \infty).$$

It follows that, point-wise in $\mathbf{v}$,

$$\begin{aligned} \hat{h}(g) &\geq \hat{h}(M_g) + \hat{h}'(M_g)(g - M_g) \\ &= \hat{h}(M_g) + \ln(M_g)(g - M_g) + g - M_g. \end{aligned}$$

By Lemma 9.3,

$$\int_{\mathbb{R}^d} \hat{h}(g) d\mathbf{v} \geq \int_{\mathbb{R}^d} \hat{h}(M_g) d\mathbf{v} + \int_{\mathbb{R}^d} \ln(M_g)(g - M_g) d\mathbf{v} + \int_{\mathbb{R}^d} (g - M_g) d\mathbf{v} = \int_{\mathbb{R}^d} \hat{h}(M_g) d\mathbf{v}.$$

This shows that,

$$h[g] \geq h[M_g] = h[M_*],$$

and the result is proven. ■

10 Space-Homogeneous Model

Definition 10.1. We say that the kinetic distributions $f_i : \mathbb{R}^3 \times [0, T] \rightarrow [0, \infty)$, $i = 1, \dots, N$, satisfy the **space-homogeneous multi-species Boltzmann equation** iff, for all $\mathbf{v} \in \mathbb{R}^3$ and $t \in (0, T]$,

$$\partial_t f_i(\mathbf{v}, t) = Q_i[f_i, \mathbf{f}](\mathbf{v}, t), \tag{126}$$

where $Q_i[f_i, \mathbf{f}]$, the **cumulative binary collision operator**, is defined as

$$Q_i[f_i, \mathbf{f}](\mathbf{v}, t) := \sum_{j=1}^N Q_{i,j}[f_i, f_j](\mathbf{v}, t). \quad (127)$$

For the space-homogeneous model, the approach of the system to the equilibrium Maxwellian states is clear.

Proposition 10.2 (H-Theorem). *Suppose that $f_1, \dots, f_N : \mathbb{R}^3 \times [0, T] \rightarrow [0, \infty)$ are kinetic distributions that satisfy the space-homogeneous multi-species Boltzmann equation. Assume that the collision kernels $B_{i,j}$ have proper structure. Then,*

$$\partial_t h(t) = \sigma_h(t) \leq 0, \quad (128)$$

where h is the **information content**, defined as

$$h(t) := \sum_{i=1}^N \int_{\mathbb{R}^3} f_i(\mathbf{v}, t) \ln(f_i(\mathbf{v}, t)) \, d\mathbf{v},$$

and σ_h is the **information loss rate**, defined via

$$\sigma_h(t) := \sum_{i=1}^N \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}, t) \ln(f_i(\mathbf{v}, t)) \, d\mathbf{v}.$$

Proof. The result follows from the proof of Proposition 7.3. ■

The following also follows from a previously-established result.

Proposition 10.3. *Suppose that $f_i : \mathbb{R}^3 \times [0, \infty) \rightarrow [0, \infty)$, $i = 1, \dots, N$, are kinetic distributions with the properties*

$$\int_{\mathbb{R}^3} f_i(\mathbf{v}, t) |\mathbf{v}|^k \, d\mathbf{v} < \infty, \quad k = 0, 1, 2, 3, \quad (129)$$

for every $t \geq 0$. Assume that the kinetic distributions satisfy the space-homogeneous multi-species Boltzmann system. Then, for all $t \geq 0$, and for each $i = 1, \dots, N$,

$$d_t \rho_i(t) = 0, \quad (130)$$

$$\rho_i d_t \mathbf{u}_i(t) = m_i \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}, t) \mathbf{v} \, d\mathbf{v}, \quad (131)$$

and

$$d_t \left(\frac{\rho_i}{2} |\mathbf{u}_i(t)|^2 + e_i(t) \right) = \frac{m_i}{2} \int_{\mathbb{R}^3} Q_i[f_i, \mathbf{f}](\mathbf{v}, t) |\mathbf{v}|^2 d\mathbf{v}. \quad (132)$$

Consequently, using the collisional invariants,

$$d_t \rho(t) = 0, \quad (133)$$

$$\rho d_t \mathbf{u}(t) = 0, \quad (134)$$

and

$$d_t \omega(t) = d_t \sum_{i=1}^N \left(\frac{\rho_i}{2} |\mathbf{u}_i(t)|^2 + e_i(t) \right) = 0. \quad (135)$$

Remark 10.4. *It can be proven that solutions to the space-homogeneous multi-species Boltzmann equation dissipate the information content,*

$$h(t) := \sum_{i=1}^N \int_{\mathbb{R}^3} f_i(\mathbf{v}, t) \ln(f_i(\mathbf{v}, t)) d\mathbf{v},$$

until the system reaches an equilibrium state (at $t = \infty$) characterized by

$$0 = [\sigma_h]_\infty := \sum_{i=1}^N \int_{\mathbb{R}^3} Q_i[f_{i,\infty}, \mathbf{f}_\infty](\mathbf{v}) \ln(f_{i,\infty}(\mathbf{v})) d\mathbf{v}.$$

The kinetic distributions evolve in such a way that ρ_i is time invariant, and the total momentum and energy are conserved. At equilibrium, the kinetic distributions will be Maxwellians, as guaranteed by Proposition 7.4. The particle temperatures will equilibrate to same temperatures, that is,

$$\lim_{t \rightarrow \infty} \theta_i(t) = \theta_\infty > 0, \quad i = 1, \dots, N.$$

11 References

- [1] P.L. Bhatnagar, E.P. Gross, and M. Krook. A model for collision processes in gases I: Small amplitude processes in charged and neutral one-component systems. *Phys. Rev.*, 94:511, 1954.
- [2] Alberto Bressan. Notes on the boltzmann equation: Lecture notes for a summer course at sissa. Technical report, Pennsylvania State University, 2005.

- [3] Carlo Cercignani. *Rarefied Gas Dynamics*. Cambridge University, 2000.
- [4] Stewart Harris. *Introduction to the Theory of the Boltzmann Equation*. Dover Publications, 2004.
- [5] Gilberto Medeiros Kremer. *An Introduction to the Boltzmann Equation and Transport Processes in Gases*. Springer, 2010.
- [6] Sandra Pieraccini and Gabriella Puppo. Implicit–explicit schemes for BGK kinetic equations. *Journal of Scientific Computing*, 32:1–28, 2007.
- [7] Marlies Pirner. *Kinetic Modelling of Gas Mixtures*. PhD thesis, University of Wuerzburg, 2018.
- [8] H. Struchtrup. *Macroscopic Transport Equations for Rarefied Gas Flows: Approximation Methods in Kinetic Theory*. Springer Verlag, Berlin, Germany, 2005.

A An Important Integral

We need to compute the following generalized Gaussian integral in several places in the report.

Lemma A.1.

$$\int_{\mathbb{R}^d} |\mathbf{s}|^2 e^{-|\mathbf{s}|^2} d\mathbf{s} = \frac{d}{2} \pi^{\frac{d}{2}}. \quad (136)$$

Proof.

$$\begin{aligned}
\int_{\mathbb{R}^d} |\mathbf{s}|^2 e^{-|\mathbf{s}|^2} d\mathbf{s} &= \int_{\mathbb{R}^d} (s_1^2 + \dots + s_d^2) e^{-(s_1^2 + \dots + s_d^2)} d\mathbf{s} \\
&= \int_{\mathbb{R}^d} \left(\sum_{j=1}^d s_j^2 \right) \prod_{k=1}^d e^{-s_k^2} d\mathbf{s} \\
&= \int_{\mathbb{R}^d} s_1^2 \prod_{k=1}^d e^{-s_k^2} d\mathbf{s} + \dots + \int_{\mathbb{R}^d} s_d^2 \prod_{k=1}^d e^{-s_k^2} d\mathbf{s} \\
&= \left(\prod_{\substack{k=1 \\ k \neq 1}}^d \int_{\mathbb{R}} e^{-s_k^2} ds_k \right) \int_{\mathbb{R}} s_1^2 e^{-s_1^2} ds_1 + \dots + \left(\prod_{\substack{k=1 \\ k \neq d}}^d \int_{\mathbb{R}} e^{-s_k^2} ds_k \right) \int_{\mathbb{R}} s_d^2 e^{-s_d^2} ds_d \\
&= \left(\pi^{\frac{1}{2}} \right)^{d-1} \int_{\mathbb{R}} s_1^2 e^{-s_1^2} ds_1 + \dots + \left(\pi^{\frac{1}{2}} \right)^{d-1} \int_{\mathbb{R}} s_d^2 e^{-s_d^2} ds_d \\
&= \pi^{\frac{d-1}{2}} \sum_{i=1}^d \int_{\mathbb{R}} s_i^2 e^{-s_i^2} ds_i, \quad (137)
\end{aligned}$$

where we have used many elementary integration techniques, and the fact that $\int_{\mathbb{R}} e^{-x^2} dx = \pi^{\frac{1}{2}}$. It remains to determine the value of the terms of the form $\int_{\mathbb{R}} s^2 e^{-s^2} ds$. To that end, utilizing integration by parts, with

$$u = s_i \implies du = ds_i \quad (138)$$

$$dv = 2s_i e^{-s_i^2} ds_i \implies v = -e^{-s_i^2}, \quad (139)$$

we have

$$\begin{aligned} \frac{1}{2} \int_{\mathbb{R}} 2s_i^2 e^{-s_i^2} ds_i &= \frac{1}{2} \left[\left[-s_i e^{-s_i^2} \right]_{-\infty}^{\infty} - \int_{\mathbb{R}} -e^{-s_i^2} ds_i \right] \\ &= \frac{1}{2} \int_{\mathbb{R}} e^{-s_i^2} ds_i \\ &= \frac{1}{2} \pi^{\frac{1}{2}}. \end{aligned} \quad (140)$$

Thus, Equation (137) gives us

$$\int_{\mathbb{R}^d} |\mathbf{s}|^2 e^{-|\mathbf{s}|^2} d\mathbf{s} = \pi^{\frac{d-1}{2}} \sum_{i=1}^d \frac{1}{2} \pi^{\frac{1}{2}} = \pi^{\frac{d-1}{2}} \left(\frac{d}{2} \pi^{\frac{1}{2}} \right) = \frac{d}{2} \pi^{\frac{d}{2}}, \quad (141)$$

as desired. ■